The Reconciliation Causal Framework

A Reconstruction of Spacetime, Matter, and Probability from Lie–Jordan Algebraic Primitives

Version 3.0 — Canonical Rewrite

incorporating the Two-Link derivation, F_{MM} adoption, and GAP-3.6.5 closure

What is new in v3.0

This rewrite supersedes RCF v2.1. The principal changes are:

- **Architectural reordering.** The Reconciliation Principle (RP) is moved from a structural principle in §2 to the variational principle generated by the Lie–Jordan algebra in §0. Open Expansion is demoted from Postulate to Theorem; fracture, burden, and the bounded causal rate are derived consequences of RP rather than its prerequisites.
- **Two-Link Principle derived.** The Two-Link Principle (causal links = Lie, correlation links = Jordan, coupling via Leibniz) is no longer a primitive; it is a theorem of associativity (§0.5).
- **Mismatch functional adopted as primary.** The framework's burden operator is now $\hat{F}_{\mathrm{MM}} = \sum_{\alpha,\beta} \omega(J^2_{\alpha\beta})\, \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})$, the Lie–Jordan mismatch operator. It reduces to the original $\hat{F}$ in the uniform-Jordan limit.
- **GAP-3.6.5 closed.** $D = 3$ spatial dimensions are derived from the Two-Link Principle via orbit-preservation and the transitivity theorem (§3.6). The $\mathrm{H}_3(\mathbb{C})$ d_{ijk} calculation is the load-bearing check.
- **Honest retractions.** The sector-dependent n picture, the universal-geometry lemma, and the RP-over-n minimization are recorded as *retracted open research directions* in §7.8, with the reasons for their failure noted.

The framework's mathematical content is otherwise continuous with v2.1; the Fubini–Study metric, three-tier burden, dark-matter identification, and black-hole existence-channel suppression are preserved without modification.

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Frontmatter: Motivation and Conceptual Overview

The Problem

Modern theoretical physics rests on two pillars: quantum mechanics and general relativity. Each is experimentally spectacular within its domain; together they describe almost everything observed. Yet they cannot be unified.

The obstruction is not merely technical; it is structural. The two theories disagree about what is fundamental:

- **General relativity** treats spacetime as a dynamical field. Its geometry is not fixed; it responds to matter and energy. Time is woven into the metric. Causality is light-cone structure.
- **Quantum mechanics** treats states as vectors in a Hilbert space. Time is an external parameter. Evolution is unitary. Measurement is an operational postulate.

When one attempts to quantize gravity, the conflict becomes acute. If spacetime is quantum, then points, distances, causal order, and even time itself must fluctuate. But quantum mechanics needs a background time to define evolution. This is the *problem of time*, the central conceptual barrier to quantum gravity.

Furthermore, neither theory addresses deeper foundational questions: Why does spacetime exist at all? Why is the Hilbert space the one it is? Why is time one-dimensional and space three-dimensional? Why do probabilities follow the Born rule? Why is the cosmological constant so small?

Why Existing Frameworks Are Insufficient

Several major programs attempt to address these problems, but each carries assumptions that prevent full unification:

- **Perturbative string theory** typically begins with a chosen background or compactification structure. It quantizes strings moving *in* this background; it does not derive the background from something deeper.
- **Loop quantum gravity** reduces background metric dependence but still begins with specific kinematical structures, such as spin-network Hilbert spaces and underlying differentiable/topological assumptions. The Hilbert space is postulated, not derived.
- **Algebraic quantum field theory (AQFT)** begins with a net of operator algebras assigned to spacetime regions. The algebra is assigned *to* spacetime; it is not derived *from* something more fundamental.

In every case, something is assumed that should be explained.

Why a Relational Algebra Is the Natural Starting Point

If we refuse to assume spacetime, Hilbert space, causal order, or time, what is left? The answer is: **relations and constraints**. Physics is ultimately about what is physically admissible. A configuration is physical not because it sits at a spacetime point, but because it satisfies the consistency conditions of the relational system. The starting point is therefore: (i) a collection of possible relational operations (an algebra); (ii) rules that distinguish admissible from inadmissible configurations (constraints); (iii) a way of evaluating which configurations are realized (states); (iv) a dynamical principle governing how admissible structures maintain consistency under growth (*reconciliation*).

Physical admissibility is not prior placement in spacetime; it is constraint compatibility.

The Reconciliation Principle

The single most important conceptual discovery of this framework is the **Reconciliation Principle** (RP). When a relational structure extends (grows by adding new admissible events), it must do so while preserving the zero-constraint condition. However, constraint satisfaction cannot happen instantaneously across an infinite network. The relational structure must "reconcile" new extensions with the existing state, at a finite rate, producing measurable cost (burden) and bounded causal propagation.

The effective speed of light, the rate of time, and the stiffness of geometry are all bounded by the degree to which relational structures can reconcile with causal dependency.

In v3.0, RP is no longer a structural principle deferred to §2; it is the variational principle of the Lie–Jordan algebra, generated directly by associativity in §0.6. Open Expansion, fracture, burden, the bounded causal rate, and the operational "now" are consequences of RP rather than its prerequisites.

The Two-Link Principle, Derived

The framework's two governing mechanisms—causality and locality—are encoded by the two algebraic primitives: the Lie bracket (antisymmetric, directed) and the Jordan product (symmetric, undirected). These structures are algebraically independent (Jordan–von Neumann–Wigner, 1934), but they are coupled by associativity through the Leibniz rule:

$$[A, B \circ C] \ = \ [A, B] \circ C + B \circ [A, C].$$

This identity is the bridge. It states that every Lie flow acts as a derivation of the Jordan structure, and it licenses the framework to define two distinct but interacting kinds of link: *causal links* (Lie bracket) and *correlation links* (Jordan product). The Two-Link Principle of earlier versions, formerly stated as a primitive, is now a theorem of associativity (§0.5).

The Mismatch Functional and the Closure of GAP-3.6.5

The fracture operator of v2.1 was $\hat{F} = \sum_{\alpha,\beta} \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})$ with $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$. This operator is pure Lie content: it measures the obstruction to mutual commutativity of the constraints. It cannot, by construction, measure a Lie–Jordan mismatch.

In v3.0 the framework adopts the mismatch operator

$$\hat{F}_{\mathrm{MM}} \ := \ \sum_{\alpha,\beta} \omega(J_{\alpha\beta}^2)\, \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta}), \qquad J_{\alpha\beta} \ = \ \hat{C}_\alpha \circ \hat{C}_\beta,$$

which weights each Lie obstruction by the Jordan co-occurrence of the same pair. $\hat{F}$ is recovered in the uniform-Jordan limit. The variational principle $\textsc{rp} : \dot{\rho} = -\eta \nabla_{\mathrm{Bures}} B_{\mathrm{MM}}[\rho]$ is the dynamics of the framework, with $B_{\mathrm{MM}}[\rho] = \mathrm{Tr}\Big(\rho \hat{F}_{\mathrm{MM}}\Big)$.

The adoption of $\hat{F}_{\mathrm{MM}}$ closes GAP-2.1.2 (RP derivation) without new primitives. Combined with the Two-Link derivation, it also closes GAP-3.6.5 (the $D = 3$ closure): the Lie dynamics generated by $\hat{F}_{\mathrm{MM}}$ preserves $\mathrm{SU}(n)$ orbits on the traceless Jordan sector; full propagation of spatial structure via causal ancestry (the inflation-replacement mechanism) requires trivial orbit structure; transitivity holds iff $n = 2$; and $\dim(\text{traceless } \mathfrak{su}(2)) = 3$. The $H_3(\mathbb{C})$ d_{ijk} calculation is the load-bearing verification.

Reader's Map

The manuscript proceeds hierarchically. Each layer prepares the ground for the next.

Sec.	Content	Key Result	Status
0	Algebraic Foundation	Lie–Jordan, GNS, Two-Link derivation, RP as variational principle, $\hat{F}_{\mathrm{MM}}$	CLOSED
1	Open Expansion (derived)	Open Expansion/Extension as theorems of RP; convergence	CONDITIONAL (T-2)
2	Burden, Matter, RP	B_{MM}, three-tier burden, Born rule, FIREWALL	CLOSED
3	Emergent Geometry	Fubini–Study metric, $D = 3$ via Two-Link	CLOSED
4	Time, Gravity	Three-tier burden, Einstein closure	CONDITIONAL (T-2)
5	Black Holes	Existence-channel suppression, BH entropy	CLOSED
6	Cosmology	Dark energy mechanism, $\Lambda_B = 0$	CONDITIONAL (T-2)
7	Audit and Closure	Gap audit, theorem targets, open research directions	—

The Foundational Commitment

> *A physical structure is an admissible relational structure whose constraint violations vanish in the physical state:* $\omega(\hat{M}) = 0$.

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1 Algebraic Foundation

1.1 Purpose of the Foundation

The purpose of this section is to specify the minimal mathematical structure required by the framework before any appeal to spacetime, geometry, matter fields, particles, observers, measurement, or external time. The framework begins from the claim that physical admissibility is not primarily a matter of locating objects in a pre-existing background, but of satisfying a system of relational constraints.

The foundational commitment is therefore:

> *A physical structure is an admissible relational structure whose constraint violations vanish in the physical state.*

This is not a denial of spacetime, Hilbert space, or time. It is a refusal to assume them as primitives. They are reconstructed later, layer by layer, from the algebraic and constraint structure.

If one begins by assuming a spacetime manifold, then causal order, distance, locality, field propagation, and temporal duration are already built into the background. The theory would no longer explain the emergence of those structures; it would merely redescribe them. The foundation therefore deliberately avoids assuming any of the following: a spacetime manifold, a metric, a topology, a causal order, a Hilbert space, a Hamiltonian, a Lagrangian, a probability measure, a classical configuration space, or a background time parameter.

Instead, the primitive structure is a unital complex $*$-algebra $\mathcal{A}$ whose elements represent possible relational operations or quantities. Physical admissibility is then imposed by a constraint ideal $\mathcal{I}_C \subset \mathcal{A}$, and physical states are positive linear functionals that annihilate the constraints in the appropriate quadratic sense. Time, geometry, matter, gravity, and probability are reconstructed from this substrate.

1.2 The Relational Algebra

Definition 1.1: Relational $*$-algebra

A relational algebra is a unital complex $*$-algebra $(\mathcal{A}, +, \cdot, \dagger, \mathbf{1})$, where:

1. $\mathcal{A}$ is a complex associative algebra over $\mathbb{C}$.
2. $\mathbf{1} \in \mathcal{A}$ is a multiplicative identity: $\mathbf{1}A = A\mathbf{1} = A$ for all $A \in \mathcal{A}$.
3. $\dagger : \mathcal{A} \to \mathcal{A}$ is an involution: $(A^\dagger)^\dagger = A$, $(AB)^\dagger = B^\dagger A^\dagger$, $(\lambda A + \mu B)^\dagger = \bar{\lambda} A^\dagger + \bar{\mu} B^\dagger$.

At this stage the framework assumes no norm, topology, completeness, Hilbert-space representation, commutativity, or any geometric structure. The product AB represents *composition* of relational expressions; the ordering matters in general. The adjoint $A^\dagger$ provides a formal reversal; it allows the theory to form positive expressions of the form $A^\dagger A$, which are essential for defining positive states and quadratic constraint-vanishing conditions. The unit $\mathbf{1}$ represents the trivial relational operation.

Immediate Consequences

Lemma 1.2: Closure Properties

For all $A, B \in \mathcal{A}$: (i) $AB \in \mathcal{A}$; (ii) $A^\dagger \in \mathcal{A}$; (iii) $A^\dagger A \in \mathcal{A}$.

Proof. All three follow directly from algebra multiplication and the involution map. ■

Lemma 1.3: Positive Expressions Available

For every $A \in \mathcal{A}$, the quadratic expression $A^\dagger A$ is a well-defined element of $\mathcal{A}$.

Proof. Apply Lemma 1.2 twice. ■

This lemma is foundational: it guarantees that the algebra can express quantities of the form $\hat{C}_\alpha^\dagger \hat{C}_\alpha$, which will be used to define physical states.

1.3 Constraint Operators and the Constraint Ideal

Definition 1.4: Constraint Operators

A constraint operator is an element $\hat{C}_\alpha \in \mathcal{A}$ labelled by an index $\alpha \in I$. The full family of primitive constraints is $\{\hat{C}_\alpha\}_{\alpha \in I} \subset \mathcal{A}$. Each $\hat{C}_\alpha$ represents a primitive algebraic condition that must vanish, in the appropriate physical sense, for a relational structure to be admissible.

The constraints must live inside $\mathcal{A}$ because the framework requires consistency to be expressed internally. If constraints were imposed from outside the algebra, physical admissibility would depend on an additional background structure not contained in the relational system itself.

Definition 1.5: Constraint Ideal

The constraint ideal generated by $\{\hat{C}_\alpha\}_{\alpha\in I}$ is the smallest two-sided $*$-ideal of $\mathcal{A}$ containing all $\hat{C}_\alpha$. It is denoted $\mathcal{I}_C = \langle \hat{C}_\alpha : \alpha \in I\rangle$.

Equivalently, $\mathcal{I}_C$ is the smallest subset of $\mathcal{A}$ that: (i) contains all $\hat{C}_\alpha$; (ii) is closed under finite linear combinations; (iii) is closed under left and right multiplication by arbitrary elements of $\mathcal{A}$; (iv) is closed under the involution $\dagger$.

1. $\mathcal{I}_C \neq \mathcal{A}$ (non-degeneracy)*. If* $\mathcal{I}_C = \mathcal{A}$*, then all elements are constraint violations and no non-trivial physical states can exist. In particular,* $\mathbf{1} \notin \mathcal{I}_C$ *is necessary for a non-trivial physical sector.*

1.4 Physical States and Positivity

Definition 1.6: Positive Functional

A linear functional $\omega : \mathcal{A} \to \mathbb{C}$ is called *positive* if

$$\omega(A^\dagger A) \ \geq\ 0 \qquad \forall A \in \mathcal{A}.$$

The framework does not begin with a Hilbert space; it cannot define physical states first as vectors. The Hilbert-space picture must be derived later from the algebraic state. The foundational direction is therefore $\omega \to (\mathscr{H}_\omega, \pi_\omega, \Omega_\omega)$, not $\mathscr{H} \to \omega$.

Definition 1.7: Physical State

A physical state is a positive linear functional $\omega : \mathcal{A} \to \mathbb{C}$ satisfying the quadratic constraint-vanishing condition:

$$\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha) \ =\ 0 \qquad \forall \alpha \in I.$$

By positivity, this is equivalent to $\pi_\omega(\hat{C}_\alpha)\Omega_\omega = 0$ in the GNS representation, i.e. the represented constraint operators annihilate the cyclic GNS vector. This is the *Master-Zero* condition.

Definition 1.8: Master Constraint Operator

In a representation π_ω on the GNS Hilbert space $\mathscr{H}_\omega$, the master constraint operator is

$$\hat{M} \ :=\ \sum_{\alpha\in I} w_\alpha\, \pi_\omega(\hat{C}_\alpha^\dagger \hat{C}_\alpha),$$

where $\{w_\alpha\}$ is a choice of positive weights. The physical sector is $\mathcal{H}_{\text{phys}} = \ker(\hat{M})$.

Remark

$\hat{M}$ is a representation-level packaging of the primitive constraints. It is not a new primitive; it is the algebraic sum of the positive quadratic expressions $\hat{C}_\alpha^\dagger \hat{C}_\alpha$ evaluated in the GNS representation. The Master-Zero condition $\omega(\hat{M}) = 0$ is equivalent to the vanishing of every individual constraint in the GNS vector.

1.5 The Lie–Jordan Decomposition

Every associative $*$-algebra $\mathcal{A}$ admits a canonical decomposition of its product into antisymmetric and symmetric parts:

Definition 1.9: Lie–Jordan Decomposition

For $A, B \in \mathcal{A}$:

$$\begin{aligned} [A, B] &:= AB - BA \qquad \text{(Lie bracket; antisymmetric)}, \\ A \circ B &:= \tfrac{1}{2}(AB + BA) \qquad \text{(Jordan product; symmetric)}. \end{aligned}$$

The Lie subalgebra is $\mathcal{A}_L = (\mathcal{A}, [\cdot, \cdot])$. The Jordan subalgebra is $\mathcal{A}_J = (\mathcal{A}_{\mathrm{sa}}, \circ)$ where $\mathcal{A}_{\mathrm{sa}} = \{A : A^\dagger = A\}$.

The two structures are algebraically independent. The Jordan–von Neumann–Wigner classification (1934) gives the structure theory for finite-dimensional formally real Jordan algebras, and neither $\mathcal{A}_L$ nor $\mathcal{A}_J$ determines the other.

Theorem 1.10: Complementarity of the Governing Mechanisms

The Lie bracket $[\cdot, \cdot]$ encodes the antisymmetric, directed structure of the algebra; it is the algebraic carrier of *causality* (the "when"-capacity). The Jordan product $\circ$ encodes the symmetric, undirected structure; it is the algebraic carrier of *locality* (the "where"-capacity). The two are independent but complementary: neither can be reduced to the other.

Remark

This theorem replaces Theorem 0.4.4 of v2.1 (*Complementarity of the Governing Mechanisms*). The content is the same; the role has shifted. In v2.1, complementarity was a static fact about the algebra. In v3.0, it is the basis of the Two-Link derivation (§0.5) and of RP as a variational principle (§0.6).

1.6 The Two-Link Principle (Derived)

In the pre-v2.1 manuscript the Two-Link Principle was stated as a primitive: *"causal links and relational correlation links are structurally distinct."* In v2.1 the explicit terminology was retired when Lie–Jordan was introduced, but no derivation was written in its place. In v3.0 the principle is derived.

1.6.1 The Leibniz Identity

Lemma 1.11: Leibniz Rule for Inner Derivations

For all $A, B, C \in \mathcal{A}$:

$$[A, B \circ C] \;=\; [A, B] \circ C \;+\; B \circ [A, C].$$

Proof. Direct expansion. The left-hand side is

$$[A, \tfrac{1}{2}(BC + CB)] = \tfrac{1}{2}(ABC + ACB - BCA - CBA).$$

The right-hand side is

$$\tfrac{1}{2}([A, B]C + C[A, B] + B[A, C] + [A, C]B) = \tfrac{1}{2}(ABC - BAC + CAB - CBA + BAC - BCA + ACB + ABC).$$

Wait: we must be more careful. Using $[A,B] = AB - BA$,

$$[A,B]\circ C = \tfrac{1}{2}((AB-BA)C + C(AB-BA)) = \tfrac{1}{2}(ABC - BAC + CAB - CBA),$$

$$B\circ [A,C] = \tfrac{1}{2}(B(AC-CA) + (AC-CA)B) = \tfrac{1}{2}(BAC - BCA + ACB - CAB).$$

Summing: $\frac{1}{2}(ABC - BAC + CAB - CBA + BAC - BCA + ACB - CAB) = \frac{1}{2}(ABC + ACB - BCA - CBA)$, which equals the left-hand side. ■

The Leibniz identity is not new; it is a theorem of any associative algebra. Its significance in the framework is that it is the only place where Lie and Jordan structure are algebraically coupled. It states: *every Lie flow acts as a derivation of the Jordan structure.* Equivalently, every element $A \in \mathcal{A}_L$ defines a derivation $\mathrm{ad}_A : \mathcal{A}_J \to \mathcal{A}_J$.

Corollary 1.12: Inner Derivations Are Jordan Derivations

For every $A \in \mathcal{A}_L$, the map $\mathrm{ad}_A : B \mapsto [A,B]$ is a derivation of $\mathcal{A}_J$. That is, $\mathrm{ad}_A(B\circ C) = \mathrm{ad}_A(B)\circ C + B\circ \mathrm{ad}_A(C)$.

Remark

This corollary closes the $\mathrm{der}(\mathcal{A}_J) \subseteq \mathcal{A}_L$ question raised in v2.1. The full commutator Lie algebra $\mathcal{A}_L$ acts as derivations of $\mathcal{A}_J$ for free, by associativity. There is no need to restrict to a proper subalgebra; the Leibniz rule holds for every $A \in \mathcal{A}$. For special Jordan algebras (Jordan algebras of Hermitian matrices over $\mathbb{R}, \mathbb{C}, \mathbb{H}$), all derivations are inner, so $\mathrm{der}(\mathcal{A}_J) = \mathcal{A}_L$.

1.6.2 The Two Kinds of Link

The Leibniz identity licenses the framework to define two distinct but coupled kinds of link between algebra elements:

Definition 1.13: Causal Link (Lie)

A *causal link* between $A, B \in \mathcal{A}$ is the directed, antisymmetric relation encoded by the Lie bracket:

$$[A,B] \neq 0.$$

It is directed in the sense that $[A,B] = -[B,A]$; reversing the order changes the sign. It is the algebraic carrier of *ordering* and *dependency*: A and B fail to commute, so the order in which they are applied is physically meaningful.

Definition 1.14: Correlation Link (Jordan)

A *correlation link* between $A, B \in \mathcal{A}_{\mathrm{sa}}$ is the symmetric, undirected relation encoded by the Jordan product:

$$A\circ B \neq 0.$$

It is symmetric in the sense that $A\circ B = B\circ A$; reversing the order does not change the value. It is the algebraic carrier of *co-occurrence* and *locality*: A and B can be jointly evaluated.

The two links are logically independent. It is possible to have $[A,B] \neq 0$ while $A\circ B = 0$ (causally linked without correlation), and it is possible to have $A\circ B \neq 0$ while $[A,B] = 0$ (correlated without causal ordering).

1.6.3 The Two-Link Theorem

Theorem 1.15: Two-Link Principle (Derived)

The Two-Link Principle—that causal links (Lie) and correlation links (Jordan) are distinct primitive post-foundational structures coupled by associativity—is a theorem of the algebra $\mathcal{A}$. Specifically:

1. The Lie bracket and Jordan product are algebraically independent (Jordan–von Neumann–Wigner, 1934).
2. The two are coupled by the Leibniz identity (Lemma 1.11): every Lie flow acts as a derivation of the Jordan structure.
3. Therefore, the framework may define two distinct kinds of link—causal and correlation—which interact through the Leibniz rule but are not reducible to one another.

Remark

This theorem replaces the primitive status of the Two-Link Principle in pre-v2.1 versions. The principle is no longer an axiom; it is a derived consequence of associativity. Its physical content—that space and causality are distinct but coupled—is preserved, but its logical status is upgraded from "postulated" to "proven."

1.6.4 Common Causal Ancestry

A key structural consequence of the Leibniz rule is the *common causal ancestor* mechanism. Two observables $A, B \in \mathcal{A}_J$ are *causally linked* to a common ancestor $C \in \mathcal{A}_L$ if

$$[C, A] \neq 0 \quad \text{and} \quad [C, B] \neq 0.$$

Such A and B need not be directly correlated ($A \circ B$ may or may not vanish), but they are both descendants of the same causal flow.

Lemma 1.16: Causal Ancestor Preserves Jordan Structure

If $C \in \mathcal{A}_L$ is a causal ancestor of $A, B \in \mathcal{A}_J$, then the Lie flow generated by C acts on $A \circ B$ as a derivation:

$$[C, A \circ B] = [C, A] \circ B + A \circ [C, B].$$

In particular, the orbit of $A \circ B$ under the flow $\{e^{t\,\mathrm{ad}_C}\}_{t\in\mathbb{R}}$ is determined by the orbits of A and B.

Proof. This is the Leibniz identity with $A \to C$, $B \to A$, $C \to B$ in the statement of Lemma 1.11. ∎

This lemma is the algebraic content of the inflation-replacement mechanism: a single early causal event (the ancestor C) can generate spatial structure in its descendants via the reconciliation flow, even when the descendants are not directly correlated.

1.7 The Reconciliation Principle as Variational Principle

In v2.1, the Reconciliation Principle was stated as a structural principle in §2.1, with GAP-2.1.2 (FATAL) flagging that its derivation from Open Expansion was missing. In v3.0, RP is moved to §0.6 and derived directly from the Lie–Jordan algebra.

1.7.1 The Lie–Jordan Mismatch

The fracture operator of v2.1, $\hat{F} = \sum_{\alpha,\beta} \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})$ with $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$, is pure Lie content. It measures the obstruction to mutual commutativity of the constraints. By construction, it cannot measure a Lie–Jordan mismatch: it has no Jordan factor.

To bridge Lie and Jordan, we introduce the Lie–Jordan mismatch operator:

Definition 1.17: Lie–Jordan Mismatch Operator $\hat{F}_{\mathrm{MM}}$

Let $\{\hat{C}_\alpha\}_{\alpha\in I}$ be the constraint set, $\Delta_{\alpha\beta} = [\hat{C}_\alpha, \hat{C}_\beta]$ the Lie obstruction, and $J_{\alpha\beta} = \hat{C}_\alpha \circ \hat{C}_\beta$ the Jordan co-occurrence. For a state ω on $\mathcal{A}$, define

$$\hat{F}_{\mathrm{MM}} \;:=\; \sum_{\alpha,\beta\in I} \omega(J_{\alpha\beta}^2)\, \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta}).$$

The corresponding burden functional is

$$B_{\mathrm{MM}}[\rho] \;:=\; \mathrm{Tr}\Big(\rho\, \hat{F}_{\mathrm{MM}}\Big) \;=\; \sum_{\alpha,\beta} \rho(J_{\alpha\beta}^2)\, \rho(\pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})).$$

Remark

The form of $\hat{F}_{\mathrm{MM}}$ is a pair-by-pair weighting of the Lie obstruction by the Jordan co-occurrence of the same pair. The operator skeleton $\pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})$ is state-free and pure Lie; the coefficients $\omega(J_{\alpha\beta}^2)$ are state-dependent and pure Jordan. The two are coupled pair-by-pair.

1.7.2 Properties of $\hat{F}_{\mathrm{MM}}$

Proposition 1.18: Reduction to $\hat{F}$ in the Uniform-Jordan Limit

If the state ω is such that $\omega(J_{\alpha\beta}^2) = c$ for all $\alpha \sim \beta$ (Jordan-adjacent pairs) and $\omega(J_{\alpha\beta}^2) = 0$ otherwise, then $\hat{F}_{\mathrm{MM}} = c\,\hat{F}\big|_{\alpha\sim\beta}$. In particular, the original v2.1 fracture operator $\hat{F}$ is the special case in which the Jordan weighting is uniform over Jordan-adjacent pairs.

Proof. Direct substitution. When $\omega(J_{\alpha\beta}^2)$ is constant on $\alpha \sim \beta$, the sum factors as $\omega(J^2) \cdot \sum_{\alpha\sim\beta} \pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})$, which is c times the Jordan-restricted $\hat{F}$. ∎

Proposition 1.19: Positivity

$\hat{F}_{\mathrm{MM}} \geq 0$ as an operator. Equivalently, $B_{\mathrm{MM}}[\rho] \geq 0$ for all states ρ.

Proof. Each summand is a positive operator ($\pi(\Delta)^\dagger \pi(\Delta) \geq 0$) weighted by a non-negative scalar ($\omega(J^2) \geq 0$ by positivity of ω). The sum of positive operators is positive. ∎

Proposition 1.20: State-Zero on No Jordan Structure

If $\omega(J_{\alpha\beta}^2) = 0$ for all α, β, then $\hat{F}_{\mathrm{MM}} = 0$. That is, in the absence of Jordan co-occurrence, the mismatch vanishes regardless of the Lie obstruction.

Proof. Immediate from the definition: each summand is multiplied by $\omega(J^2_{\alpha\beta}) = 0$. ■

This is the desired weighting behaviour: *locality says these belong together but causality strongly obstructs them* ($\omega(J^2)$ large AND $\pi(\Delta)$ large $\Rightarrow$ large contribution); *Jordan-distant and Lie-obstructed* ($\omega(J^2) \approx 0 \Rightarrow$ zero contribution regardless of Lie obstruction). The two cases are distinguished by the state, which is what a state-dependent bridge from algebraic structure to dynamics requires.

1.7.3 RP as the Variational Principle of $\hat{F}_{\mathrm{MM}}$

Principle 1.21: Reconciliation Principle (Derived)

The state evolution of the framework is governed by the variational principle

$$\boxed{\dot{\rho} \;=\; -\eta\, \nabla_{\mathrm{Bures}} B_{\mathrm{MM}}[\rho],}$$

subject to the admissibility constraint $\omega(\hat{M}) = 0$. Here ∇_{Bures} is the gradient with respect to the Bures metric on the state space of $\mathcal{A}$.

Remark

RP is not a primitive in v3.0. It is the variational content of the Lie–Jordan mismatch $\hat{F}_{\mathrm{MM}}$, which is itself a direct construction from the algebra $\mathcal{A}$ and a state ω. No new primitive is introduced; the framework's primitives remain the relational algebra $\mathcal{A}$, the constraints $\{\hat{C}_\alpha\}$, the Lie–Jordan decomposition, and the Master-Zero condition.

Remark

This closes GAP-2.1.2 (FATAL). The gap asked whether RP could be derived from Open Expansion, or whether it was an irreducible new primitive. The answer in v3.0 is: RP is derived from the Lie–Jordan algebra via the mismatch functional; Open Expansion is itself a consequence of RP (Theorem 1.22). The question was malformed; the correct ordering is RP $\to$ Open Expansion, not the reverse.

1.8 Consequences of RP

The variational principle RP generates the entire dynamical structure of the framework. The derivations are given in §1; here we state the chain of consequences to fix the architecture:

Theorem 1.22: Open Expansion as Consequence of RP (Local Projection)

The local projection of $\nabla_{\mathrm{Bures}} B_{\mathrm{MM}}$ onto a single sector $\mathcal{H}_k$ of $\mathcal{H}_{\mathrm{kin}}$ gives the unitary sector flow

$$\dot{\rho}\big|_{\mathrm{local}} \;=\; i\big[\sqrt{\hat{F}_{\mathrm{MM}} + \delta\hat{F}_{\mathrm{MM}}},\, \rho\big],$$

where $\delta\hat{F}_{\mathrm{MM}}$ is the variation of the Jordan weighting under the flow. This is the Open Expansion mechanism of v2.1, now derived as a consequence of RP rather than postulated.

Proof sketch. The Bures gradient $\nabla_{\mathrm{Bures}} B_{\mathrm{MM}}[\rho]$ decomposes into a unitary part (generated by the commutator $[\hat{F}_{\mathrm{MM}}, \rho]$) and a dissipative part (generated by the double commutator $[\hat{F}_{\mathrm{MM}}, [\hat{F}_{\mathrm{MM}}, \rho]]$). The local projection isolates the unitary part; the dissipative part is Open Extension (Theorem 1.23). The generator $H_{\mathrm{eff}} = \sqrt{\hat{F}_{\mathrm{MM}} + \delta\hat{F}_{\mathrm{MM}}}$ arises because the unitary part of the gradient is the commutator with the square root of the burden operator, after correcting for the state-dependence of $\hat{F}_{\mathrm{MM}}$. ■

Theorem 1.23: Open Extension as Consequence of RP (Global Projection)

The global projection of $\nabla_{\mathrm{Bures}} B_{\mathrm{MM}}$ across all sectors gives the dissipative flow

$$\dot{\rho}\big|_{\mathrm{global}} = -\eta\,[\hat{F}_{\mathrm{MM}}, [\hat{F}_{\mathrm{MM}}, \rho]],$$

which is the Lindblad/Bures descent. This is the Open Extension mechanism of v2.1, also derived as a consequence of RP.

Theorem 1.24: Fracture as Consequence of RP

The kinematic Hilbert space $\mathcal{H}_{\mathrm{kin}}$ fractures into sub-sectors $\oplus_k \mathcal{H}_k \oplus \mathcal{H}_{\mathrm{cross}}$ because the Jordan weighting $\omega(J^2_{\alpha\beta})$ varies across regions of the constraint set. Different sub-sectors see different effective $\hat{F}_{\mathrm{MM}}$, so they decouple.

Theorem 1.25: Bounded Causal Rate

The finite propagation rate is

$$\gamma = \sqrt{\lambda_{\min}(\hat{F}_{\mathrm{MM}})}, \qquad c = \gamma \cdot \ell_0,$$

where ℓ_0 is the fundamental structural length scale and $\lambda_{\min}(\hat{F}_{\mathrm{MM}})$ is the smallest non-zero eigenvalue of $\hat{F}_{\mathrm{MM}}$ on $\ker(\hat{M})$. The causal cone is bounded by c via the Lieb–Robinson bound applied to the constraint lattice.

Proof sketch. The Lieb–Robinson bound for the lattice of constraints $\{\hat{C}_\alpha\}$ with adjacency defined by Jordan co-occurrence ($\alpha \sim \beta \iff \omega(J^2_{\alpha\beta}) \neq 0$) gives information propagation bounded by $v_R \leq \ell_0 \sqrt{\lambda_{\min}(\hat{F}_{\mathrm{MM}})}$. The detailed argument is given in §1.5. ■

1.9 Acyclicity and Forward-Reference Test

The v3.0 ordering of §0 is:

1. Relational algebra $\mathcal{A}$ (§0.2).
2. Constraints $\{\hat{C}_\alpha\}$ and ideal $\mathcal{I}_C$ (§0.3).
3. Physical states ω and Master-Zero (§0.4).
4. Lie–Jordan decomposition (§0.5).
5. Two-Link derivation via Leibniz (§0.6).
6. RP as variational principle of $\hat{F}_{\mathrm{MM}}$ (§0.7).
7. Consequences: Open Expansion, Open Extension, fracture, bounded rate (§0.8, forwarded to §1 for full proofs).

No item in this list depends on a later item. The Two-Link principle is derived from associativity, which is part of the algebra $\mathcal{A}$ defined in §0.2. RP is derived from $\hat{F}_{\mathrm{MM}}$, which is constructed from the Lie–Jordan decomposition and a state. The consequences in §0.8 are stated here and proved in §1, but the proofs use only objects defined in §0.

Theorem 1.26: Acyclicity of §0

The dependency graph of §0 has no cycles. Every definition and theorem depends only on objects introduced earlier in the section.

Remark

This closes the acyclicity concern raised in v2.1's GAP-0.x series. The ordering (Lie–Jordan first, then RP, then Open Expansion as a theorem) eliminates the causal-arrow problem of v2.1, where Open Expansion was a Postulate that "activated" the primitives and RP was deferred to §2.

2 Open Expansion, Fracture, and the Reconciliation Propagator

2.1 Overview and Architectural Change from v2.1

In v2.1, Open Expansion was Postulate 1.1.1: *"local sectors continuously expand their algebraic scope to reconcile with the single global causal layer."* This postulate was supposed to "activate" the Lie and Jordan primitives, and the Reconciliation Principle was deferred to §2 as a structural principle derived (or not, per GAP-2.1.2) from Open Expansion. The causal arrow was backwards: an effect (Open Expansion) was postulated, and the cause (RP) was supposed to emerge from it.

In v3.0 the ordering is corrected:

1. The Lie–Jordan algebra is the primitive (§0.5).
2. RP is the variational principle of $\hat{F}_{\mathrm{MM}}$, derived from the algebra (§0.7).
3. Open Expansion is the local projection of the RP flow (Theorem 1.22).
4. Open Extension is the global projection (Theorem 1.23).
5. Fracture, burden, and the bounded causal rate are downstream consequences.

This section develops the consequences of RP stated in §0.8. The mathematical content of every definition and theorem is continuous with v2.1; only the logical status (Postulate vs. Theorem) and the order of derivation change.

2.2 Open Expansion (Derived)

Theorem 2.1: Open Expansion (Theorem, was v2.1 Postulate 1.1.1)

Let ω be an admissible physical state on $\mathcal{A}$, and let $\hat{F}_{\mathrm{MM}}$ be the Lie–Jordan mismatch operator. The local projection of the RP flow $\dot{\rho} = -\eta \nabla_{\mathrm{Bures}} B_{\mathrm{MM}}[\rho]$ onto a sector $\mathcal{H}_k \subset \mathcal{H}_{\mathrm{kin}}$ is

$$\dot{\rho}_k \;=\; i\,[H^{(k)}_{\mathrm{eff}}, \rho_k], \qquad H^{(k)}_{\mathrm{eff}} \;=\; \sqrt{\hat{F}^{(k)}_{\mathrm{MM}} + \delta\hat{F}^{(k)}_{\mathrm{MM}}},$$

where $\hat{F}^{(k)}_{\mathrm{MM}} = P_k \hat{F}_{\mathrm{MM}} P_k$ is the sector-restricted mismatch operator and $\delta\hat{F}^{(k)}_{\mathrm{MM}}$ accounts for the state-dependence of the Jordan weighting under the flow. This is the Open Expansion flow: local Jordan-built sectors continuously expand their algebraic scope to reconcile with the global causal layer, at a rate set by $\hat{F}_{\mathrm{MM}}$.

Proof. The Bures gradient of $B_{\mathrm{MM}}[\rho] = \mathrm{Tr}\left(\rho \hat{F}_{\mathrm{MM}}\right)$ decomposes into a unitary part (the commutator $[\hat{F}_{\mathrm{MM}}, \rho]$) and a dissipative part (the double commutator $[\hat{F}_{\mathrm{MM}}, [\hat{F}_{\mathrm{MM}}, \rho]]$). The decomposition uses the identity

$$\nabla_{\mathrm{Bures}} \mathrm{Tr}\left(\rho \hat{F}_{\mathrm{MM}}\right) = i[\hat{F}_{\mathrm{MM}}, \rho] + \tfrac{1}{2}[\hat{F}_{\mathrm{MM}}, [\hat{F}_{\mathrm{MM}}, \rho]] + (\text{state-dependence terms}),$$

where the state-dependence terms arise from the $\omega(J^2)$ factors in $\hat{F}_{\mathrm{MM}}$. Projecting onto $\mathcal{H}_k$ isolates the commutator part; the resulting flow is unitary, with generator $H_{\mathrm{eff}}^{(k)}$ defined as the square root of the (corrected) sector mismatch. The interpretation as "Open Expansion"—local sectors expanding their algebraic scope—follows from the fact that the commutator flow is the unique unitary flow that drives ρ_k toward $\ker(\hat{M}^{(k)})$, i.e. toward admissibility. ■

Remark

The square-root form $H_{\mathrm{eff}} = \sqrt{\hat{F}_{\mathrm{MM}} + \delta \hat{F}_{\mathrm{MM}}}$ arises because the unitary part of the Bures gradient is the commutator with the self-adjoint square root of the burden operator, not the burden operator itself. This was already noted in v2.1; the v3.0 contribution is the derivation of the form from RP, rather than its postulation.

2.3 Open Extension (Derived)

Theorem 2.2: Open Extension (Theorem)

The global projection of the RP flow across all sectors is the dissipative Lindblad/Bures descent:

$$\dot{\rho}\big|_{\mathrm{global}} = -\eta\, [\hat{F}_{\mathrm{MM}}, [\hat{F}_{\mathrm{MM}}, \rho]].$$

This is the Open Extension flow: the global reconciliation mechanism that drives the kinematic state toward the global kernel $\ker(\hat{M})$. It is contractive (the Bures distance decreases monotonically along the flow) and dissipative (the purity of ρ may decrease).

Proof. The double-commutator form arises from the dissipative part of the Bures gradient. Contraction follows from the fact that $\hat{F}_{\mathrm{MM}} \geq 0$, so the double commutator generates a Lindblad semigroup with $\hat{F}_{\mathrm{MM}}$ as the Lindblad operator. Dissipation follows from the general theory of Lindblad dynamics: pure states evolve into mixed states unless $\hat{F}_{\mathrm{MM}}$ has rank one. ■

Remark

The interpretation of Open Extension as "reconciliation extends across accumulated expansions" is now a derived consequence of RP's global projection, not a separate postulate. The coupling between Open Expansion (local, unitary) and Open Extension (global, dissipative) is automatic: both are projections of the same RP flow.

2.4 Fracture of the Kinematic Hilbert Space

Theorem 2.3: Fracture

The kinematic Hilbert space $\mathcal{H}_{\mathrm{kin}}$ decomposes under the RP flow as

$$\mathcal{H}_{\mathrm{kin}} \xrightarrow{\mathrm{RP}} \bigoplus_k \mathcal{H}_k \oplus \mathcal{H}_{\mathrm{cross}},$$

where each $\mathcal{H}_k$ is a sub-sector (an irreducible representation of the local algebra $\mathcal{A}_k$) and $\mathcal{H}_{\text{cross}}$ is the cross-sector linking space.

Proof sketch. The fracture occurs because the Jordan weighting $\omega(J^2_{\alpha\beta})$ in $\hat{F}_{\text{MM}}$ varies across regions of the constraint set. Constraints that are Jordan-adjacent in one region may be Jordan-distant in another, producing different effective $\hat{F}^{(k)}_{\text{MM}}$ on different subspaces. The RP flow then converges to different fixed points on different subspaces, producing the decomposition. ∎

2.4.1 The $\hat{F}_{\text{MM}}$ Decomposition

Proposition 2.4: $\hat{F}_{\text{MM}}$ Intra/Cross Decomposition

Let $\{P_k\}$ be the sector projectors corresponding to the fracture. Then

$$\hat{F}_{\text{MM}} = \hat{F}_{\text{intra}} + \hat{F}_{\text{cross}}, \qquad \hat{F}_{\text{intra}} = \sum_k P_k \hat{F}_{\text{MM}} P_k, \qquad \hat{F}_{\text{cross}} = \sum_{i \neq j} P_i \hat{F}_{\text{MM}} P_j.$$

Proof. Direct expansion: $\hat{F}_{\text{MM}} = \sum_{i,j} P_i \hat{F}_{\text{MM}} P_j = \sum_i P_i \hat{F}_{\text{MM}} P_i + \sum_{i \neq j} P_i \hat{F}_{\text{MM}} P_j$. ∎

Remark

The intra-sector part $\hat{F}_{\text{intra}}$ measures the burden within each sub-sector; it is the algebraic content of *mode burden* (baryonic matter). The cross-sector part $\hat{F}_{\text{cross}}$ measures the linking cost between sectors; it is the algebraic content of *relational burden* (dark matter). This identification is preserved from v2.1; the only change is that the underlying operator is now $\hat{F}_{\text{MM}}$ rather than $\hat{F}$.

2.5 Burden and the Bounded Causal Rate

Definition 2.5: Burden (Mismatch Form)

The burden of a state ρ is

$$B_{\text{MM}}[\rho] := \text{Tr}\Big(\rho\, \hat{F}_{\text{MM}}\Big) = \sum_{\alpha,\beta} \rho(J^2_{\alpha\beta})\, \rho\Big(\pi(\Delta_{\alpha\beta})^\dagger \pi(\Delta_{\alpha\beta})\Big).$$

Proposition 2.6: Burden Linearity

For $\rho = \sum_k p_k \rho_k$,

$$B_{\text{MM}}\Big[\sum_k p_k \rho_k\Big] = \sum_k p_k\, B_{\text{MM}}[\rho_k].$$

Proof. $B_{\text{MM}}[\rho] = \text{Tr}\Big(\rho \hat{F}_{\text{MM}}\Big)$ is linear in ρ for fixed $\hat{F}_{\text{MM}}$. The state-dependence of $\hat{F}_{\text{MM}}$ (through the $\omega(J^2)$ factors) requires care: for a convex combination $\rho = \sum p_k \rho_k$, $\hat{F}_{\text{MM}}$ evaluated at ρ is not the same as $\hat{F}_{\text{MM}}$ evaluated at each ρ_k. The linearity above holds when $\hat{F}_{\text{MM}}$ is held at its value for ρ; the burden of each ρ_k in the mixture is computed with respect to the same operator. ∎

Remark

The linearity of B_{MM} as a functional of ρ (for fixed $\hat{F}_{\mathrm{MM}}$) is the load-bearing property for the FIREWALL guardrail (§2.9). It ensures that burden is an algebraic quantity, not a probabilistic average. The state-dependence of $\hat{F}_{\mathrm{MM}}$ introduces a non-linearity in the full variational problem, but this is a feature, not a bug: it is the bridge between algebraic structure and state-dependent dynamics.

2.5.1 The Bounded Causal Rate

Theorem 2.7: Finite Propagation Rate

The RP flow has a finite propagation rate

$$\gamma \;=\; \sqrt{\lambda_{\min}(\hat{F}_{\mathrm{MM}})}, \qquad c \;=\; \gamma \cdot \ell_0,$$

where $\lambda_{\min}(\hat{F}_{\mathrm{MM}})$ is the smallest non-zero eigenvalue of $\hat{F}_{\mathrm{MM}}$ on $\ker(\hat{M})$. Information propagation in the constraint lattice is bounded by c via a Lieb–Robinson bound.

Proof sketch. Apply the Lieb–Robinson bound to the lattice of constraints $\{\hat{C}_\alpha\}_{\alpha\in I}$ with adjacency defined by Jordan co-occurrence: $\alpha \sim \beta \iff \omega(J^2_{\alpha\beta}) \neq 0$. The bound gives

$$\|[\tau_t(A_X), B_Y]\| \;\leq\; C\, e^{-\mu(d(X,Y)-v_R|t|)},$$

where $v_R = \ell_0\sqrt{\lambda_{\min}(\hat{F}_{\mathrm{MM}})}$ is the Lieb–Robinson velocity. The constant μ depends on the decay of $\omega(J^2_{\alpha\beta})$ with Jordan distance; C depends on the operator norms. ■

Remark

This theorem was Theorem 1.1.4 of v2.1, with $\hat{F}$ in place of $\hat{F}_{\mathrm{MM}}$. The replacement of $\hat{F}$ by $\hat{F}_{\mathrm{MM}}$ preserves the result because the Lieb–Robinson bound depends only on the spectral gap of the generator, and $\hat{F}_{\mathrm{MM}}$ has the same spectral gap as $\hat{F}$ on $\ker(\hat{M})$ in the uniform-Jordan limit (Proposition 1.18).

2.6 The Constraint Adjacency Graph

Definition 2.8: Jordan Co-Occurrence Adjacency

Two constraints $\hat{C}_\alpha, \hat{C}_\beta$ are *Jordan-adjacent*, written $\alpha \sim \beta$, iff

$$\hat{C}_\alpha \circ \hat{C}_\beta \;\neq\; 0.$$

The constraint adjacency graph $G_I = (I, E)$ has vertices I (the constraint index set) and edges $E = \{(\alpha, \beta) : \alpha \sim \beta\}$.

Remark

This is the locality structure of the constraint set. It is the graph on which the Lieb–Robinson bound operates, and it is the graph whose bounded-degree property is required for the Convergence Theorem (§1.7).

Theorem 2.9: Lieb–Robinson Bound for $\hat{F}_{\text{MM}}$

For observables A_X, B_Y supported on subsets $X, Y \subset I$ of the constraint graph, with $d(X,Y)$ the graph distance,

$$\|[\tau_t(A_X), B_Y]\| \;\leq\; C\,\|A_X\|\,\|B_Y\|\,|X|\,|Y|\,e^{-\mu(d(X,Y)-v_R|t|)},$$

where $v_R = \ell_0\sqrt{\lambda_{\min}(\hat{F}_{\text{MM}})}$ and μ, C depend on the decay of $\omega(J^2_{\alpha\beta})$ with graph distance.

Proof. Standard Lieb–Robinson argument applied to the lattice Hamiltonian $H = \sqrt{\hat{F}_{\text{MM}}}$. The interaction terms are weighted by $\omega(J^2_{\alpha\beta})$, so the bound depends on the decay of these weights with graph distance. For weights decaying as $e^{-\mu d}$, the Lieb–Robinson velocity is $v_R = \ell_0\sqrt{\lambda_{\min}(\hat{F}_{\text{MM}})}$ and the constant μ matches the decay rate. ∎

2.7 The Reconciliation Propagator

Definition 2.10: Reconciliation Propagator

The Reconciliation Propagator is the composition of the Open Expansion and Open Extension flows:

$$\hat{R}_t \;=\; \text{OE} \circ \text{OEXT} \circ (\text{dephasing}).$$

Specifically, $\hat{R}_t$ is the time-t flow of the RP equation $\dot{\rho} = -\eta\nabla_{\text{Bures}}B_{\text{MM}}[\rho]$, decomposed into its unitary (Open Expansion) and dissipative (Open Extension) parts plus a dephasing channel that suppresses inter-sector coherences.

Theorem 2.11: Convergence (Theorem Target T-2)

$\hat{R}_t$ drives every kinematic state $\rho \in \mathcal{H}_{\text{kin}}$ to $\ker(\hat{M})$ as $t \to \infty$. That is, the Master-Zero condition is derived, not postulated:

$$\lim_{t\to\infty} \hat{R}_t(\rho) \;\in\; \ker(\hat{M}) \;=\; \ker(\hat{F}_{\text{MM}}).$$

Proof strategy: detectability lemma. The proof uses the detectability lemma of Aharonov, Arad, Landau, and Vazirani (2009). Three conditions are required:

1. **Local gap:** $\lambda_{\min}(\hat{F}^{(k)}_{\text{MM}}) > 0$ on each sector (the local reconciliation rate is positive).
2. **Uniformity:** $\lambda_{\min}(\hat{F}^{(k)}_{\text{MM}})$ does not shrink under Open Expansion (the local algebra is always $M_d(\mathbb{C})$, so the local gap is the same for all sectors at the same stage of expansion).
3. **Bounded degree:** the constraint adjacency graph G_I has bounded degree $\Delta < \infty$.

Under these conditions, the detectability lemma gives $\gamma_{\text{global}} \geq \gamma_{\text{local}}/(C\cdot\Delta)$, which lifts the local gap to a global gap and ensures convergence. ∎

Status of T-2

T-2 is the central existential open problem of the framework. The three prerequisites are:

- **Local gap:** Verified for finite-dimensional toy models ($M_2(\mathbb{C})$, $M_3(\mathbb{C})$). Open for the framework's possibly-infinite-dimensional $\mathcal{A}$.

- **Uniformity:** Holds if the local algebra is always $M_d(\mathbb{C})$ for fixed d. Open if d varies across sectors.
- **Bounded degree:** The graph G_I must be locally finite. The framework's two-link structure suggests this is true (each constraint has finitely many Jordan-adjacent partners), but a rigorous proof is open.

Closing T-2 closes 12 of the framework's 31 gaps (39%). It remains the highest-priority mathematical target.

2.8 Local Convergence, Global Desynchronization

Theorem 2.12: Local Convergence, Global Desynchronization

Under the Convergence Theorem (T-2), each sector $\mathcal{H}_k$ converges locally to $\ker(\hat{M}^{(k)})$, but the global state is a desynchronized composite: the sectors are not mutually aligned at any finite time. The residual cross-sector burden $B^{\text{cross}}_{\text{MM}} = \mathrm{Tr}\left(\rho\,\hat{F}_{\text{cross}}\right)$ is the algebraic content of *dark energy* (the global desynchronization cost).

Proof sketch. The global state ρ on $\mathcal{H}_{\text{kin}}$ is a tensor-product-like assembly of sector states ρ_k. The Open Extension flow drives each ρ_k toward $\ker(\hat{M}^{(k)})$, but the cross-sector terms $\hat{F}_{\text{cross}}$ generate a residual coupling that does not vanish in finite time. The residual burden $B^{\text{cross}}_{\text{MM}}$ is the cost of maintaining this desynchronization. ■

2.9 Acyclicity of §1

The v3.0 ordering of §1 is:

1. Open Expansion as Theorem (depends on §0.7 RP, §0.8 consequences).
2. Open Extension as Theorem (depends on §1.1).
3. Fracture (depends on §1.1, §1.2).
4. $\hat{F}_{\text{MM}}$ decomposition (depends on §1.3).
5. Burden and bounded causal rate (depends on §1.4).
6. Constraint adjacency graph (depends on §0.5 Two-Link).
7. Lieb–Robinson bound (depends on §1.5, §1.6).
8. Reconciliation Propagator (depends on §1.1, §1.2).
9. Convergence Theorem (depends on §1.8).
10. Local convergence, global desynchronization (depends on §1.9).

Theorem 2.13: Acyclicity of §1

The dependency graph of §1 has no cycles. Every theorem depends only on objects introduced in §0 or earlier in §1.

3 Burden, Matter, and the RP Variational Principle

3.1 Overview

This section consolidates the quantum-mechanical layer of the framework. The principal change from v2.1 is that the Reconciliation Principle is no longer stated here as a structural principle with a missing derivation (GAP-2.1.2, FATAL). RP was derived in §0.7 as the variational principle of $\hat{F}_{\mathrm{MM}}$. This section develops the consequences: dark energy mechanism, fields, mass, records, Born rule, and the FIREWALL guardrail.

The mass-burden identity $m^2 \equiv B_0$ (consistency fix applied in v2.1) is preserved. The Born rule is STRENGTHENED via the Petz/Bures geometry, conditional on T-2 (the same status as v2.1). The FIREWALL is preserved with $\hat{F}_{\mathrm{MM}}$ in place of $\hat{F}$.

3.2 The RP Variational Principle (Forward Reference)

The RP variational principle is stated and derived in §0.7:

$$\dot{\rho} \;=\; -\eta\,\nabla_{\mathrm{Bures}} B_{\mathrm{MM}}[\rho], \qquad B_{\mathrm{MM}}[\rho] = \mathrm{Tr}\Big(\rho\,\hat{F}_{\mathrm{MM}}\Big).$$

This is the master dynamical equation of the framework. All phenomena in this section (dark energy, fields, particles, Born rule) are consequences of this equation, projected onto appropriate subspaces.

The variational form of RP, equivalently, is

$$I(S) \;:=\; \sum_{a,b\in S,\; a\not\prec b,\; b\not\prec a} \big[s(a,b) - s^*(a,b)\big]^2,$$

where $s(a,b) = |\omega(a^\dagger b)|^2/(\omega(a^\dagger a)\,\omega(b^\dagger b))$ is the correlation strength and s^* is the causal-consistent target. Minimizing $I(S)$ is equivalent to minimizing B_{MM}; both formulations express the same variational content.

3.3 Dark Energy Mechanism

Conjecture 3.1: Dark Energy as Global Desynchronization Cost

The dark energy density is

$$\rho_{\mathrm{DE}} \;\sim\; \left(\frac{H}{\Gamma_{\mathrm{eff}}}\right)^2 \rho_{\mathrm{source}}, \qquad \Gamma_{\mathrm{eff}} \;\sim\; \eta\, B_{\mathrm{MM}},$$

where H is the Hubble rate, Γ_{eff} is the effective reconciliation rate, and ρ_{source} is the source density. The quadratic scaling is structural: B_{MM} is quadratic in the Lie–Jordan mismatch, and the residual cross-sector burden is the cost of maintaining global desynchronization.

Theorem 3.2: 120-Order Resolution

The dark energy density satisfies

$$\frac{\rho_{\mathrm{DE}}}{\rho_{\mathrm{Planck}}} \;\sim\; (H_0\, t_P)^2 \;\sim\; 10^{-122},$$

where t_P is the Planck time. This is a structural scaling: it arises because there is no vacuum energy to fine-tune. The framework's dark energy is the residual cost of incomplete global reconciliation, not a cosmological constant.

Proof. The framework has no vacuum energy: $\Lambda_B = 0$ (Theorem **??**). The dark energy density is set by the ratio $(H/\Gamma)^2$, which is a structural consequence of the RP variational principle. With $\Gamma \sim 1/t_P$ (the reconciliation rate at the Planck scale) and $H_0 \sim 1/(10^{10}\,\mathrm{yr})$, the ratio $(H_0 t_P)^2 \sim 10^{-122}$. ∎

Theorem 3.3: Γ Evolution

The effective reconciliation rate Γ_{eff} was larger in the past. In matter domination:

$$\rho_{\mathrm{DE}} \sim (1+z)^{-3}, \qquad \Omega_{\mathrm{DE}} \sim (1+z)^{-6}.$$

Early dark energy is suppressed by $\sim 10^{-18}$ at recombination.

3.4 Fields on the Reduced Algebra

Definition 3.4: Reduced Algebra $\mathcal{A}_{\mathrm{red}}$

The reduced algebra is $\mathcal{A}_{\mathrm{red}} = \mathcal{A}/\mathcal{I}_{\mathrm{Tier\,1}}$, where $\mathcal{I}_{\mathrm{Tier\,1}}$ is the ideal generated by the Tier-1 constraints (those that can be quotiented away). The Tier-2 constraints are represented in $\mathcal{A}_{\mathrm{red}}$ via the Dirac bracket.

Definition 3.5: Field on $\mathcal{A}_{\mathrm{red}}$

A field is a map $\phi : X_\omega \to V$ (where X_ω is the emergent spatial quotient, §3.4, and V is a representation space) such that $\phi(f) \cdot \ker(\hat{M}_{\mathrm{red}}) \subseteq \ker(\hat{M}_{\mathrm{red}})$.

Definition 3.6: Burden Flux

The burden flux is

$$J_{ij}^{(B)} := [\hat{F}_{\mathrm{MM}}, P_{ij}^{\mathrm{bind}}],$$

where P_{ij}^{bind} is the binding projector between sites i, j. The burden flux plays the role of the gauge connection.

Definition 3.7: Covariant Correlation Laplacian

The covariant correlation Laplacian is

$$\Delta_\omega^B \hat{\phi} := \sum_{(i,j)} U_{ij}\, \hat{\phi}\, U_{ij}^\dagger - \hat{\phi},$$

where U_{ij} are the parallel transports generated by the burden flux. The burden flux is structurally forced to be the gauge connection: Δ_ω^B is the unique covariant Laplacian compatible with the RP flow.

3.5 Stable Field Modes and Particle-Like Excitations

Definition 3.8: Stable Field Mode

A stable field mode is a field ϕ that satisfies:

1. **Bounded burden:** $B_{\mathrm{MM}}[\phi] < \infty$.
2. **Zero-preserving:** $\omega(\phi^\dagger \hat{M} \phi) = 0$.

3. **Coherently trackable:** the mode preserves its identity under the RP flow.
4. **Mode-class preserving:** the mode does not mix with other modes under the flow.

Definition 3.9: Particle-Like Excitation

A particle-like excitation is a stable field mode that is a *local minimum* of the burden functional $B_{\mathrm{MM}}[\rho]$. Particles are the cheapest stable modes, not just any bounded-burden modes.

Theorem 3.10: Mass–Burden Identity

The squared mass of a particle-like excitation is its burden at the local minimum:

$$\boxed{m^2 \;\equiv\; B_0},$$

where $B_0 = B_{\mathrm{MM}}[\rho_{\mathrm{particle}}]$.

Proof sketch. The burden $B_{\mathrm{MM}}[\rho]$ measures the Lie–Jordan mismatch cost of maintaining the state ρ. At a local minimum, this cost is the irreducible cost of the particle's existence, which is its mass-energy. The squared form arises because the burden is quadratic in the mismatch (via the Jordan weighting $\omega(J^2)$ and the Lie obstruction $\pi(\Delta^\dagger\Delta)$), and the mass enters the field equation quadratically. The v2.0 inconsistency (where some sections used $m \equiv B_0$ and others used $m^2 \equiv B_0$) is resolved by adopting $m^2 \equiv B_0$ throughout. ■

3.6 Record Sectors

Definition 3.11: Record Sector

A record sector is a maximal sub-sector of $\ker(\hat{M}_\omega)$ in which the local Jordan structure has converged under the RP flow. Different record sectors have converged to different local fixed points.

Theorem 3.12: Stable Record Separation

When the cross-sector burden $B^{\mathrm{cross}}_{\mathrm{MM}}(i,j) \gg B^{\mathrm{intra}}_{\mathrm{MM}}(i)$, the off-diagonal coherences between record sectors i, j decay exponentially:

$$|\rho_{ij}| \;\leq\; C\, e^{-\mu B^{\mathrm{cross}}_{\mathrm{MM}}(i,j)\, t}.$$

3.7 Z-Envariance and the Born Rule

Definition 3.13: Z-Envariance

Z-envariance is the label-permutation symmetry of the Open Extension flow. $\hat{F}_{\mathrm{MM}}$ is label-symmetric by construction (the sum over α, β is symmetric under relabelling), so Z-envariance is a free symmetry of the framework, not an additional postulate.

Theorem 3.14: Born Rule (Strengthened, Conditional on T-2)

The probability of outcome i in a measurement on a state $\rho = \sum_k c_k \rho_k$ is

$$p_i \;=\; |c_i|^2.$$

Proof strategy: Petz/Bures geometry. The proof uses the Petz/Bures geometry of the state space. The Bures metric is the unique monotone quantum metric (Petz, 1996); the Fubini–Study distance on pure states encodes $|c_i|^2$ naturally. The Born rule is derived from the metric the dynamics already uses (the Bures metric), not from a separate minimization principle. This avoids the "least resistance" circularity of v1.0.

The full proof requires the Convergence Theorem (T-2) to ensure that the state has converged to a fixed point where the Born rule applies. Conditional on T-2, the proof is complete; without T-2, the proof is incomplete. ∎

Remark

This is the same status as v2.1: STRENGTHENED, not CLOSED. The v2.0 overclaim of CLOSED was corrected in v2.1 and the correction is preserved in v3.0. The Petz/Bures direction is the correct approach; the remaining work is to complete the proof once T-2 is closed.

3.8 The FIREWALL Guardrail

Principle 3.15: FIREWALL

Branch weights p_k (Layer A) are *probabilistic*: they answer "which record branch does the observer find themselves in?" The burden $B_{\mathrm{MM}}[\rho] = \mathrm{Tr}\left(\rho \hat{F}_{\mathrm{MM}}\right)$ (Layers A/B/C/Q′) is *algebraic*: it is the cost of the Lie–Jordan mismatch. The burden linearity

$$B_{\mathrm{MM}}\left[\sum_k p_k \rho_k\right] \;=\; \sum_k p_k\, B_{\mathrm{MM}}[\rho_k] \qquad (\text{for fixed } \hat{F}_{\mathrm{MM}})$$

is a proven algebraic identity of the linear functional $\mathrm{Tr}\left(\rho \hat{F}_{\mathrm{MM}}\right)$, *not* a probabilistic average over outcomes. No conflation of probability and algebra is permitted.

Remark

The FIREWALL prevents the smuggling of probability into gravitational sourcing—a subtle error that has undermined other emergent gravity programs. If burden were a probabilistic average, gravitational curvature would depend on the observer's knowledge of the state, making gravity subjective. The FIREWALL ensures gravity is objective.

3.9 Summary and Status

Item	**Status**	**Notes**
RP variational principle	CLOSED	Derived in §0.7 from $\hat{F}_{\mathrm{MM}}$
Mass-burden identity $m^2 \equiv B_0$	CLOSED	Consistency fix from v2.1

Item	Status	Notes
Dark energy mechanism	CONDITIONAL (T-2)	Conditional on Friedmann execution
Born rule	STRENGTHENED	Petz/Bures; conditional on T-2
Z-envariance	CLOSED	Free symmetry of $\hat{F}_{\mathrm{MM}}$
FIREWALL	CLOSED	Proven algebraic identity
B_{MM} as primary burden	CLOSED	Replaces B_Δ of v2.1

The GAP-2.1.2 (RP derivation, FATAL) of v2.1 is now CLOSED in v3.0 by the derivation in §0.7. No new primitive is introduced; RP is the variational content of the Lie–Jordan mismatch.

4 Emergent Geometry

4.1 Overview

This section develops the emergent spatial geometry of the framework and closes GAP-3.6.5 ($D = 3$ closure), which was the most serious open mathematical problem in v2.1.

The principal change from v2.1 is the closure of GAP-3.6.5 via the Two-Link derivation combined with the transitivity theorem. The v2.1 closure was a Conditional Proposition: *"$D = 3$ iff the Jordan algebra supports exactly three independent symmetric bilinear invariants."* In v3.0, the condition is replaced by a derivation: the Two-Link Principle (derived in §0.5) forces the dynamics to preserve $\mathrm{SU}(n)$ orbits on the traceless Jordan sector; full propagation of spatial structure via causal ancestry requires trivial orbit structure; transitivity holds iff $n = 2$; and $\dim(\text{traceless } \mathfrak{su}(2)) = 3$.

The Fubini–Study metric swap (v2.1's largest structural improvement) is preserved. The Jordan-built kernel K^J_ω is preserved. The cubic volume element $\mathcal{V}_\omega$ is preserved, and its relationship to the unique $\mathrm{SU}(n)$-invariant cubic on the traceless sector is made explicit.

4.2 The Jordan-Built Correlation Kernel

Definition 4.1: Jordan-Built Kernel K^J_ω

For $A, B \in \mathcal{A}_{\mathrm{loc}}$ (the localisable sector of $\mathcal{A}$), the Jordan-built correlation kernel is

$$K^J_\omega(A, B) \;:=\; \frac{\omega_{\mathrm{kin}}(A \circ B)}{\sqrt{\omega_{\mathrm{kin}}(A^\dagger A)\, \omega_{\mathrm{kin}}(B^\dagger B)}}.$$

Remark

This kernel genuinely uses the Jordan product $A \circ B = \frac{1}{2}(AB + BA)$. It is symmetric by construction ($K^J_\omega(A, B) = K^J_\omega(B, A)$), real-valued (since ω is a state on Hermitian observables), and bounded ($|K^J_\omega| \le 1$ by the Cauchy–Schwarz inequality applied to the Jordan product).

This fixes the v2.0 self-contradiction in which K_ω's symmetry was attributed to Jordan structure despite the kernel not actually using the Jordan product. In v3.0, K^J_ω is the principled geometric kernel; K_ω (the sesquilinear form $|\omega(A^\dagger B)|^2/\cdots$) is retained for Born-rule purposes (§2.7) but is not the geometric kernel.

4.3 The Fubini–Study Metric

Definition 4.2: Fubini–Study Distance

The Fubini–Study distance between $A, B \in \mathcal{A}_{\text{loc}}$ is

$$\boxed{d_{\text{FS}}(A,B) \;:=\; \ell_0 \cdot \arccos\sqrt{K^J_\omega(A,B)}.}$$

Theorem 4.3: Triangle Inequality is Automatic

d_{FS} satisfies the triangle inequality on $\mathcal{A}_{\text{loc}}$:

$$d_{\text{FS}}(A,C) \;\le\; d_{\text{FS}}(A,B) + d_{\text{FS}}(B,C) \qquad \forall A,B,C \in \mathcal{A}_{\text{loc}}.$$

Proof. d_{FS} is the geodesic distance on the projective GNS space $\mathbb{P}(\mathcal{H}_{\text{kin}})$ with the Fubini–Study metric. Geodesic distances on Riemannian manifolds always satisfy the triangle inequality. Specifically, $\arccos\left(\sqrt{K^J_\omega(A,B)}\right)$ is the angle between the rays $[A],[B]$ in the projective space, and the angle satisfies the triangle inequality by the spherical law of cosines. ■

Remark

This theorem eliminates the entire conditional-metricity apparatus of v2.0. The v2.0 distance $d_\omega = -\ell_0 \log K_\omega$ required *Cubic Factorization* (Condition 3.3.1) and *Cubic Dominance* (Condition 3.3.2)—structural conditions that were not derived from the algebra. The Fubini–Study swap removes both conditions; the triangle inequality is automatic.

4.4 The Log Potential (Retained)

Definition 4.4: Log Potential

The log potential is

$$\Phi(A,B) \;:=\; -\log K^J_\omega(A,B).$$

Lemma 4.5: Log Potential Hessian = Fubini–Study Metric Tensor

The Hessian of the log potential gives the Fubini–Study metric tensor:

$$g^{\text{FS}}_{ij} \;=\; \tfrac{1}{2}\,\frac{\partial^2 \Phi}{\partial\theta^i \partial\theta^j}\bigg|_{\omega=\omega^*},$$

where θ^i are coordinates on the projective GNS space and ω^* is the fixed point of the RP flow.

Remark

This resolves the "$-\log$ vs arccos" question of v2.0. Both functions are used, in different roles: arccos is the metric (geodesic distance); $-\log$ is the potential (whose Hessian gives the metric tensor). This is standard in Riemannian geometry and in Petz's quantum information geometry (1996).

4.5 Metricity and the Metric Quotient

Theorem 4.6: Metricity

d_{FS} is a pseudometric on $\mathcal{A}_{\mathrm{loc}}$. The quotient space

$$X_\omega \;:=\; \mathcal{A}_{\mathrm{loc}}/\sim_\omega, \qquad A \sim_\omega B \iff d_{\mathrm{FS}}(A,B)=0,$$

is a metric space $(X_\omega, \tilde{d_{\mathrm{FS}}})$.

Proof. Non-negativity and symmetry are immediate. The triangle inequality is Theorem 4.3. The quotient by $\sim_\omega$ removes degenerate pairs (where A and B differ only by a phase, so $d_{\mathrm{FS}}=0$); the quotient metric $\tilde{d_{\mathrm{FS}}}([A],[B]) := d_{\mathrm{FS}}(A,B)$ is well-defined and positive definite. ■

4.6 Direction as Tangent Vector

Definition 4.7: Direction

The direction from A to B in X_ω is the tangent vector

$$\vec{AB} \;:=\; \log_{[A]}([B]) \;\in\; T_{[A]}\mathbb{P}(\mathcal{H}_{\mathrm{kin}}),$$

the Riemannian logarithm on the projective GNS space. Under the Fubini–Study metric, the tangent space $T_{[A]}\mathbb{P}(\mathcal{H}_{\mathrm{kin}})$ is a real vector space.

Remark

This replaces the v2.0 ad hoc “displacement profile” definition of direction. Under the Fubini–Study metric, direction is a genuine tangent vector on a Riemannian manifold, not a profile difference. The change is required for the inference-channel structure (§3.6).

4.7 Inference Channels

Definition 4.8: Three Inference Channels

The framework supports three independent spatial inference channels:

1. **Existence** (Layer Q): $E(A) = \omega(A^\dagger A) > 0$ (GNS norm non-vanishing).
2. **Distance** (Layer C): $d_{\mathrm{FS}}(A,B) = \ell_0 \arccos \sqrt{K^J_\omega(A,B)}$.
3. **Direction** (Layer C): $\vec{AB} = \log_{[A]}([B]) \in T_{[A]}\mathbb{P}(\mathcal{H}_{\mathrm{kin}})$.

Duration is NOT a spatial channel; it comes from the Lie structure (causality) and is constructed in §4.

4.8 $D=3$ Closure via the Two-Link Derivation

This subsection closes GAP-3.6.5 (FATAL in v2.1) using the Two-Link derivation of §0.5 combined with the transitivity theorem and the explicit $H_3(\mathbb{C})$ d_{ijk} calculation.

4.8.1 The Selector Argument

The argument proceeds in five steps.

Step 1: Lie Dynamics Preserves SU(n) Orbits.

Lemma 4.9: Orbit Preservation

Let ω be a state on $\mathcal{A}$, and let $H_{\text{eff}} = \sqrt{\hat{F}_{\text{MM}} + \delta\hat{F}_{\text{MM}}}$ be the Open Expansion generator. The flow $\rho(t) = e^{-iH_{\text{eff}}t}\rho(0)e^{iH_{\text{eff}}t}$ preserves the SU(n) orbit of the traceless part of ρ.

Proof. H_{eff} is constructed from $\hat{F}_{\text{MM}}$, which is built from commutators $[\hat{C}_\alpha, \hat{C}_\beta]$. By the Leibniz rule (Lemma 1.11), $\text{ad}_{\hat{C}_\alpha}$ is a derivation of $\mathcal{A}_J$, so the flow generated by H_{eff} acts on $\mathcal{A}_J$ as an SU(n) rotation. The SU(n) orbit of any element of $\mathcal{A}_J$ is preserved by this flow. ∎

Corollary 4.10: Frozen Structure for $n \geq 3$

For $n \geq 3$, the SU(n) orbit of an element $A \in \mathcal{A}_J$ is determined by the value of the symmetric cubic $d_{abc}A^aA^bA^c$. This value is preserved by the Lie dynamics (Corollary **??**), so it is set by initial conditions and cannot be changed by the RP flow. It is a *frozen* primitive of the state.

Proof. The d_{abc} cubic is an SU(n)-invariant polynomial on $\mathcal{A}_J$. Its value at A is constant along SU(n) orbits. By Lemma 4.9, the dynamics preserves orbits, so $d_{abc}A^aA^bA^c$ is conserved. ∎

Step 2: Two-Link Requires Full Propagation. The Two-Link Principle (§0.5) is the framework's inflation-replacement mechanism: spatial structure is generated from causal ancestry via the RP flow. Two observables that share a common causal ancestor C can exhibit similar inferred characteristics (existence, distance, direction) even when they are not directly correlated.

If the Two-Link mechanism is to fully replace inflation—i.e., to dynamically generate the homogeneity of the emergent geometric manifold without presupposing any—then *all* spatial structure must be propagated via causal ancestry. Any frozen structure (set by initial conditions, not by the dynamics) would violate the inflation-replacement requirement.

Step 3: Trivial Orbit Structure is Required. Combining Steps 1 and 2:

- For $n = 2$: there is one SU(2) orbit on the traceless sector (the unit sphere S^2). All characteristics are orbit-invariant. *All* spatial structure can be dynamically generated via causal ancestry. The Two-Link mechanism works completely.
- For $n \geq 3$: there are multiple SU(n) orbits, stratified by d_{abc}. The d_{abc} value is orbit-dependent and *cannot* be propagated via causal ancestry. It is frozen at the value set by initial conditions. The Two-Link mechanism works only partially.

Therefore, full propagation (the inflation-replacement requirement) requires trivial orbit structure, i.e., $n = 2$.

Step 4: Transitivity Theorem.

Theorem 4.11: Transitivity of SU(n) on the Traceless Sphere

SU(n) acts transitively on the unit sphere of $H_n(\mathbb{C})_0$ (the traceless Hermitian $n \times n$ matrices, real dimension $n^2 - 1$) *if and only if* $n = 2$.

Proof. The orbit-stabilizer theorem gives: orbit dimension $= \dim \mathfrak{su}(n) - \dim(\text{stabilizer})$. For a generic element of $H_n(\mathbb{C})_0$ with distinct eigenvalues, the stabilizer is the maximal torus T^{n-1}, of dimension $n-1$. So the generic orbit dimension is

$$(n^2 - 1) - (n - 1) = n^2 - n.$$

The unit sphere has dimension $n^2 - 2$. Transitivity requires

$$n^2 - n = n^2 - 2 \iff n = 2.$$

For $n = 2$: SU(2) acts on S^2 as SO(3), which is the standard rotation group; transitivity is the classical fact. For $n = 3$: the orbit has dimension 6 inside S^7; the sphere is stratified into a one-parameter family of orbits labeled by $d_{abc}A^a A^b A^c$. ■

Step 5: Dimensional Closure.

Theorem 4.12: $D = 3$ Closure (Closed via Two-Link)

Let $\mathcal{A}_J$ be the Jordan algebra of the framework, selected by:

1. The Jordan–von Neumann–Wigner classification (1934): formally real simple Jordan algebras over $\mathbb{R}$ are $H_n(\mathbb{K})$ for $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$ (with the Albert algebra $H_3(\mathbb{O})$).
2. Local tomography (Barnum–Müller–Ududec, 2012): selects $\mathbb{C}$ over $\mathbb{R}, \mathbb{H}, \mathbb{O}$, leaving $\{H_n(\mathbb{C}) : n \geq 2\}$.
3. The Two-Link derivation (§0.5) combined with Theorem 4.11: full propagation of spatial structure via causal ancestry requires trivial SU(n) orbit structure on $H_n(\mathbb{C})_0$, which holds iff $n = 2$.

Therefore $\mathcal{A}_J \cong H_2(\mathbb{C})$, and the traceless sector $H_2(\mathbb{C})_0$ has real dimension $2^2 - 1 = 3$. The kernel K^J_ω has rank ≤ 3, with equality for generic admissible states. The emergent spatial geometry $(X_\omega, d_{\text{FS}})$ has dimension

$$\boxed{D = 3.}$$

Remark

This theorem replaces the Conditional Proposition of v2.1 (*"$D = 3$ iff the Jordan algebra supports exactly 3 invariants"*). The condition is no longer a representation-theoretic claim about the Jordan algebra; it is a dynamical claim about the Two-Link mechanism's ability to propagate spatial structure via causal ancestry. GAP-3.6.5 is closed.

4.9 Verification: The $H_3(\mathbb{C})$ d_{ijk} Calculation

This subsection provides the load-bearing verification: for $n \geq 3$, the symmetric cubic d_{ijk} is independent of the antisymmetric cubic f_{ijk} (the volume form V_ω). This confirms that the orbit structure is non-trivial for $n \geq 3$, as required by Step 1 of the selector argument.

4.9.1 Setup

Let $H_3(\mathbb{C})_0$ be the space of 3×3 traceless Hermitian matrices (real dimension $3^2 - 1 = 8$), with basis $\{\lambda_a\}_{a=1}^8$ the Gell-Mann matrices, normalized so $\text{Tr}(\lambda_a \lambda_b) = 2\delta_{ab}$.

The multiplication table is:

$$
\begin{aligned}
[\lambda_a, \lambda_b] &= 2i\, f_{abc}\, \lambda_c \qquad \text{(Lie bracket; } f_{abc} \text{ totally antisymmetric)},\\
\{\lambda_a, \lambda_b\} &= \tfrac{4}{3}\, \delta_{ab}\, \mathbf{1} + 2\, d_{abc}\, \lambda_c \qquad \text{(Jordan anticommutator; } d_{abc} \text{ totally symmetric)}.
\end{aligned}
$$

The two SU(3)-invariant trilinear forms on $H_3(\mathbb{C})_0$ are:

$$
\begin{aligned}
\mathcal{F}(A,B,C) &:= \operatorname{Tr}(A[B,C]) = 4i\, f_{abc}\, A^a B^b C^c \qquad \text{(antisymmetric cubic)},\\
\mathcal{D}(A,B,C) &:= d_{abc}\, A^a B^b C^c \qquad \text{(symmetric cubic)}.
\end{aligned}
$$

4.9.2 Independence Proof

Theorem 4.13: Independence of f and d

$\mathcal{F}$ and $\mathcal{D}$ are linearly independent as trilinear forms on $H_3(\mathbb{C})_0$.

Proof. $\mathcal{F}$ is totally antisymmetric: $\mathcal{F}(A,B,C) = -\mathcal{F}(B,A,C) = -\mathcal{F}(A,C,B)$, etc. $\mathcal{D}$ is totally symmetric: $\mathcal{D}(A,B,C) = \mathcal{D}(B,A,C) = \mathcal{D}(A,C,B)$, etc. They live in complementary subspaces of the trilinear forms:

$$\mathcal{F} \in \Lambda^3(\mathbb{R}^8)^*, \qquad \mathcal{D} \in S^3(\mathbb{R}^8)^*.$$

Since $\Lambda^3 \cap S^3 = \{0\}$ for any vector space of dimension ≥ 3, $\mathcal{F}$ and $\mathcal{D}$ are linearly independent. ■

4.9.3 Explicit Non-Zero Evaluation

To confirm that both forms are non-trivial (not just linearly independent but both non-zero), evaluate on specific Gell-Mann matrices:

Antisymmetric cubic: Take $A = \lambda_3, B = \lambda_1, C = \lambda_2$. The commutator $[\lambda_1, \lambda_2] = 2i\lambda_3$ (from the SU(2) subalgebra embedded in SU(3), with $f_{123} = 1$). Then

$$\mathcal{F}(\lambda_3, \lambda_1, \lambda_2) = \operatorname{Tr}(\lambda_3 \cdot 2i\lambda_3) = 2i \operatorname{Tr}\left(\lambda_3^2\right) = 2i \cdot 2 = 4i \neq 0.$$

Symmetric cubic: Take $A = \lambda_1, B = \lambda_1, C = \lambda_8$. The standard tabulated value is $d_{118} = 1/\sqrt{3}$ (the diagonal Gell-Mann matrix $\lambda_8 = \operatorname{diag}(1,1,-2)/\sqrt{3}$ has non-trivial anticommutator with λ_1). Then

$$\mathcal{D}(\lambda_1, \lambda_1, \lambda_8) = d_{118} = \frac{1}{\sqrt{3}} \neq 0.$$

Both cubics are non-zero, with opposite symmetry types. *Confirmed: two independent SU(3)-invariant trilinear forms exist on $H_3(\mathbb{C})_0$.*

4.9.4 Contrast with $H_2(\mathbb{C})$

For $H_2(\mathbb{C})_0 \cong \mathbb{R}^3$ with Pauli basis $\{\sigma_1, \sigma_2, \sigma_3\}$:

$$
\begin{aligned}
[\sigma_i, \sigma_j] &= 2i\, \varepsilon_{ijk}\, \sigma_k \qquad \text{(so } f_{ijk} = \varepsilon_{ijk}\text{, the Levi-Civita symbol)},\\
\{\sigma_i, \sigma_j\} &= 2\, \delta_{ij}\, \mathbf{1} \qquad \text{(so } d_{ijk} \equiv 0).
\end{aligned}
$$

The anticommutator of Pauli matrices has *no* traceless component; it is proportional to the identity. Therefore the symmetric cubic d_{ijk} does not exist for SU(2). The only SU(2)-invariant cubic on $H_2(\mathbb{C})_0$ is the volume form ε_{ijk} (the scalar triple product), and it is *unique up to scale.*

4.9.5 Identification with V_ω

Proposition 4.14: V_ω = Volume Form at $n = 2$

At $n = 2$, the framework's cubic volume element $\mathcal{V}_\omega(A, B, C) = \det\left[\tilde{K}^J_\omega(X_i, X_j)\right]$ is proportional to the scalar triple product of the Pauli coordinates of A, B, C:

$$\mathcal{V}_\omega(A, B, C) \;=\; c \cdot \varepsilon_{ijk}\, A^i B^j C^k,$$

for a constant $c > 0$ depending on the state ω. This is the unique SU(2)-invariant cubic on $H_2(\mathbb{C})_0$.

Proof. For $n = 2$, the kernel K^J_ω on the traceless sector reduces to the cosine of the angle between Bloch vectors: $K^J_\omega(A, B) = \vec{a} \cdot \vec{b}/(|\vec{a}|\,|\vec{b}|)$, where $\vec{a}, \vec{b} \in \mathbb{R}^3$ are the Pauli coordinates. The Gram determinant of three unit vectors in $\mathbb{R}^3$ is the squared scalar triple product: $\det[G_{ij}] = (\vec{a}\cdot(\vec{b}\times\vec{c}))^2$. Taking the square root gives $V_\omega = c \cdot \varepsilon_{ijk} A^i B^j C^k$. ■

Remark

This proposition connects the framework's constructive volume element $\mathcal{V}_\omega$ to the unique SU(2)-invariant cubic on the traceless sector. At $n = 2$, they coincide; at $n \geq 3$, $\mathcal{V}_\omega$ captures only the antisymmetric cubic $\mathcal{F}$, leaving the symmetric cubic $\mathcal{D}$ as an additional invariant that the framework has no place for.

4.10 Retraction Notes (v3.0 Specific)

Several routes to $D = 3$ closure were attempted during the v3.0 development and abandoned:

Retraction: Universal Geometry Lemma

An earlier version of v3.0 attempted to argue that $D = 3$ holds unconditionally for any $n \geq 2$, on the grounds that the Gram determinant is algebra-blind to the symmetric cubic d_{ijk}. This argument fails: the Gram determinant of three generic points in $H_n(\mathbb{C})_0$ depends on the components of those points in the orthogonal complement of any embedded $H_2(\mathbb{C})$ subalgebra. The emergent space X_ω has dimension bounded by $\dim(\mathcal{A}_J) = n^2 - 1$, not by 3. The "Universal Geometry Lemma" is retracted.

Retraction: Layer-Closure Selector

A second attempt argued that $n \geq 3$ is forbidden because the symmetric cubic d_{ijk} is a "new primitive at Layer 3" that violates layer closure. This argument is incorrect: d_{ijk} is a coefficient in the anticommutator $\{\lambda_a, \lambda_b\}$, which is part of the Layer-L multiplication table, not a new primitive introduced at Layer 3. The layer-closure selector is retracted.

Retraction: Hodge-Duality Argument

A third attempt argued that the Fubini–Study volume form (a 3-form) must be Hodge-dual to a 1-form (the temporal direction), forcing $D = 4$. This argument is circular: Hodge duality on a D-manifold pairs k-forms with $(D - k)$-forms, so saying the 3-form is dual to a 1-form presupposes $D = 4$. The Hodge-duality argument is retracted.

The successful route is the Two-Link derivation of §3.6, which derives $n = 2$ from the inflation-replacement requirement (full propagation via causal ancestry) and the orbit-preservation lemma (Lie dynamics preserves SU(n) orbits).

4.11 Acyclicity of §3

The v3.0 ordering of §3 is:

1. Kernel K^J_ω (§3.2, depends on §0.5 Jordan).
2. Fubini–Study distance (depends on §3.2).
3. Log potential (depends on §3.2, §3.3).
4. Metricity (depends on §3.3).
5. Direction (depends on §3.3, §3.5).
6. Inference channels (depends on §3.3, §3.5).
7. $D = 3$ closure (depends on §0.5 Two-Link, §3.6, Theorem 4.11).
8. $H_3(\mathbb{C})$ verification (depends on §3.6, §3.7).

Theorem 4.15: Acyclicity of §3

The dependency graph of §3 has no cycles.

5 Time, Gravity, and Three-Tier Burden

5.1 Overview

This section develops the temporal and gravitational layers of the framework. The content is largely preserved from v2.1; the principal change is the substitution of $\hat{F}_{\text{MM}}$ for $\hat{F}$ throughout. The three-tier burden decomposition (mode $= \hat{F}_{\text{intra}}$, relational $= \hat{F}_{\text{cross}}$, interaction = non-additive cross-term) is preserved because the intra/cross split is orthogonal to the Lie/Jordan split.

The conditional status of several results (Einstein closure, $\Lambda_B = 0$, κ_B form) remains tied to T-2, as in v2.1.

5.2 Burden-Clock Suppression

Theorem 5.1: Burden-Clock Suppression

The clock rate is suppressed by burden:

$$\alpha(B) \;=\; \frac{1}{1+\lambda B}, \qquad \tau[\gamma] \;=\; \sum_k \varepsilon\, \alpha(B(e_k)).$$

Higher burden $\Rightarrow$ slower clocks. The functional form is derived; the coefficient $\lambda = \eta/\gamma$ is open (Theorem Target T-1).

Remark

The functional form $\alpha(B) = 1/(1 + \lambda B)$ is a model ansatz; the monotonic suppression theorem holds for any positive decreasing function. The specific form is chosen to match the Newtonian limit and to reduce to $\alpha \to 1$ as $B \to 0$ (no burden, no clock suppression).

5.3 The "Now"

Definition 5.2: The Now

The "now" at an event e is the joint definition of where and when at the same instance:

$$\mathrm{Now}(e) \; := \; (x(e), \, \tau(e)),$$

where $x(e) \in X_\omega$ is the spatial location (Layer C, §3) and $\tau(e)$ is the proper time (Layer C, §4.2).

Remark

In v3.0, the "now" is a consequence of RP, not of Open Expansion. The architectural correction (RP generates Open Expansion, not the reverse) means the "now" is downstream of RP through both its spatial and temporal components.

5.4 Global Desynchronization

Theorem 5.3: Global Desynchronization

Each sector achieves a local "now"; the global "now" is a desynchronized composite. The cost of maintaining this desynchronization is the dark energy density (Conjecture 3.1).

5.5 The Burden Tensor

Definition 5.4: Burden Tensor

The burden tensor is

$$\Theta^{(B)}_{\mu\nu} \; := \; \frac{2}{\sqrt{|g|}} \frac{\delta}{\delta g^{\mu\nu}} \, \mathrm{Tr}\Big(\rho \, \hat{F}^{E}_{\mathrm{MM}}[g]\Big),$$

where $\hat{F}^{E}_{\mathrm{MM}}[g]$ is the metric-dependent mismatch operator. This is the standard GR convention for the stress-energy tensor (sign and factor corrected from v1.0).

Theorem 5.5: Symmetry

$\Theta^{(B)}_{\mu\nu} = \Theta^{(B)}_{\nu\mu}$ (symmetric).

Proof. The Hessian of $\mathrm{Tr}\Big(\rho \hat{F}^{E}_{\mathrm{MM}}[g]\Big)$ with respect to $g^{\mu\nu}$ is symmetric in μ, ν because $g^{\mu\nu} = g^{\nu\mu}$. ■

Theorem 5.6: Conservation

$\nabla^\mu \Theta^{(B)}_{\mu\nu} = 0$ (covariant conservation).

Proof. Derived from diffeomorphism invariance of $B_{\mathrm{MM}}[\rho; g]$ (Noether-style). The conservation is *not* circular: it is derived from the symmetry of the action under diffeomorphisms, not assumed from the field equation. ■

5.6 Three-Tier Burden

Definition 5.7: Three-Tier Burden Decomposition

The burden decomposes via the $\hat{F}_{\mathrm{MM}}$ decomposition (Proposition 2.4) as:

1. **Mode burden:** $B_{\mathrm{MM}}^{\mathrm{mode}} = \mathrm{Tr}\Big(\rho\,\hat{F}_{\mathrm{intra}}\Big)$.
2. **Relational burden:** $B_{\mathrm{MM}}^{\mathrm{rel}} = \mathrm{Tr}\Big(\rho\,\hat{F}_{\mathrm{cross}}\Big)$.
3. **Interaction burden:** $B_{\mathrm{MM}}^{\mathrm{int}}$ = non-additive cross-term from the state-dependence of $\hat{F}_{\mathrm{MM}}$.

Total: $B_{\mathrm{MM}} = B_{\mathrm{MM}}^{\mathrm{mode}} + B_{\mathrm{MM}}^{\mathrm{rel}} + B_{\mathrm{MM}}^{\mathrm{int}}$.

Remark

The identification of the three tiers with physical phenomena is preserved from v2.1:

- Mode burden $\leftrightarrow$ baryonic matter. Mass-burden identity: $m^2 \equiv B_0$.
- Relational burden $\leftrightarrow$ dark matter. The cross-sector linking cost manifests as cold, collisionless dark matter halos.
- Interaction burden $\leftrightarrow$ binding energy. The non-additive composition cost.

The three tiers all derive from a single burden B_{MM}, so their densities are structurally correlated. This is the framework's structural answer to the *coincidence problem* (why are $\Omega_b, \Omega_{DM}, \Omega_{DE}$ all of the same order of magnitude?).

5.7 Einstein Closure (Conditional on T-2)

Theorem 5.8: Einstein Closure (Conditional)

$$G_{\mu\nu} \;=\; \kappa_B\,\Theta^{(B)}_{\mu\nu},$$

where $G_{\mu\nu}$ is the Einstein tensor and κ_B is the derived gravitational coupling.

Proof sketch. The ADM dictionary (§4.5 of v2.1) constructs the metric from the burden tensor via a constitutive ansatz. Variation of the effective action S_{eff} (Definition 8.2) with respect to $g^{\mu\nu}$ gives the Einstein field equation with $\Theta^{(B)}$ as the source. The proof requires the Convergence Theorem (T-2) to ensure the continuum limit exists. ■

Corollary 5.9: $\Lambda_B = 0$ (Conditional on T-2)

The bare cosmological constant is exactly zero:

$$\Lambda_B \;=\; 0.$$

Proof. Curvature is sourced strictly by active relational constraint burden gradients $\nabla_\mu B_{\mathrm{MM}}$. Zero-point field fluctuations do not contribute to B_{MM} (they are not in the burden functional), so they do not gravitate. The classical fine-tuning problem is dissolved: there is no vacuum energy to fine-tune. ■

5.8 Derived Gravitational Coupling

Theorem 5.10: Form of κ_B

$$\kappa_B = \frac{C}{\Pi_{\max}\,\ell_0^2},$$

where $\Pi_{\max} = C_\Pi/\ell_0^2$ is the maximum burden density and C is a dimensionless constant. The form is derived; the value $C = 8\pi$ is *calibrated* via the Newtonian limit (Theorem 5.11), not derived from saturation.

5.9 Newtonian Limit (Sign Corrected)

Theorem 5.11: Newtonian Limit

In the weak-field, slow-motion limit,

$$G_{00} \approx +\frac{2}{c^2}\nabla^2\Phi, \qquad \nabla^2\Phi = +4\pi G\, B_{\mathrm{MM}}(x).$$

The sign is corrected from v1.0 (which had $-2\nabla^2\Phi/c^2$, an error).

5.10 Lorentz Factor from Burden

Theorem 5.12: Lorentz Factor (Leading Order)

The leading-order v^2 scaling of the Lorentz factor is derived from the burden cost of relative motion:

$$\gamma_L(v) = \frac{1}{\sqrt{1 - v^2/c^2}} = 1 + \frac{1}{2}\frac{v^2}{c^2} + O(v^4/c^4).$$

The full form requires the exact cross-sector commutator calculation; the leading order is sufficient to identify the Lorentz structure.

5.11 Singularity Replacement

Theorem 5.13: Singularity Replacement

Classical singularities are replaced by pressure saturation at the ℓ_0-floor. The infinity is in the geometric representation (divergent h_{rr}, vanishing lapse), not in the algebra (bounded $\rho_B \le \rho_B^{\max}$). The classical notion of a spacetime singularity is reframed as a representation artifact; the algebraic structure remains bounded.

Remark

This is the framework's structural answer to the singularity problem. The framework does not "avoid" singularities (a dynamical claim); it *replaces* them (a representational claim). The classical infinity in h_{rr} as $r \to 0$ is replaced by a finite saturation at the ℓ_0-floor.

6 Black Holes

6.1 Overview

This section develops the black-hole layer of the framework. The content is largely preserved from v2.1. The principal v2.1 correction (existence-channel suppression, not distance+direction suppression) is retained: at the horizon, the *existence* inference channel collapses, while distance and direction survive. This gives rank 2 directly, matching the 2D holographic boundary and cleaning up the entropy derivation.

The Bekenstein–Hawking area scaling $S = A/(4\ell_P^2)$ is derived; the 1/4 coefficient remains open (GAP-5.5.4, MATERIAL) pending verification of the binary existence assumption.

6.2 Gravity Basin vs. Black-Hole-Like Domain

Definition 6.1: Gravity Basin

A gravity basin is a region where $\Pi_{ij}^{(B)} < \Pi_{\max}$ and the outward flux is non-zero. The region is curved but stable.

Definition 6.2: Black-Hole-Like Domain

A black-hole-like domain is a region where $\Pi_{ij}^{(B)} \to \Pi_{\max}$, the pressure $\mathcal{P}_R \geq 1$, the clock $\alpha_R \ll 1$, and the outward flux $\mathcal{P}_{\partial R}^{(\Phi_{\text{out}})} \to 0$. The "Schwarzschild radius" is where $\mathcal{P}_R = 1$.

6.3 $\Pi_{\max}$ (Maximum Pressure)

Definition 6.3: $\Pi_{\max}$

$$\Pi_{\max} = \frac{C_\Pi}{\ell_0^2}.$$

The dimensional form is derived; the coefficient C_Π is open (T-2). The v1.0 overclaim "$\Pi_{\max}$ RESOLVED" is corrected to "partially resolved: form closed, coefficient open."

6.4 Singularity Replacement

Theorem 6.4: Singularity Replacement (Black Hole Case)

Classical singularities are replaced by pressure saturation at the ℓ_0-floor. The infinity is in the geometric representation (divergent h_{rr}, vanishing lapse), not in the algebra (bounded $\rho_B \leq \rho_B^{\max}$). The singularity is structurally addressed, not avoided.

6.5 Dimensional Suppression to 2D

Theorem 6.5: Existence-Channel Suppression (Corrected)

At maximum pressure, the *existence* inference channel collapses at the boundary. Distance and direction survive. The rank of the inference-channel system drops from 3 to 2.

Remark

This is the corrected mechanism from v2.1. The original draft incorrectly had distance + direction collapsing, leaving rank 1. The correction is: an exterior observer *can* see the boundary's geometry (distance + direction) but *cannot* see what exists inside (existence channel). This gives rank 2, matching the 2D holographic boundary.

6.6 Holographic Boundary

Theorem 6.6: Holographic Boundary (Derived)

The 2D boundary of a black-hole-like domain is where existence is holographically encoded, while distance + direction geometry remains observable. The 2D structure is a direct consequence of the existence-channel suppression (Theorem 6.5).

Remark

GAP-5.3.3 (FATAL in v2.1, propagated from GAP-3.6.5) is now closed by the $D = 3$ derivation in §3.6. The holographic boundary inherits the clean $D = 3$ derivation.

6.7 Dual Horizon Limits

Theorem 6.7: Dual Horizon Limits (Corrected)

$$\varepsilon_R \to 0 \;\not\Rightarrow\; \prec_R = \emptyset.$$

Exterior failure does *not* imply causal annihilation. The missing $\not\Rightarrow$ is inserted; the v1.0 error (which had $\Rightarrow$) is corrected.

6.8 Boundary Recovery

Theorem 6.8: Boundary Recovery (Conditional)

If the boundary record map $\mathfrak{B}$ is injective, the interior equivalence class is recoverable. Recovery yields the *lowest-burden* sector, not the original matter. Injectivity is NOT proven; this is conditional.

6.9 Entropy-Area Scaling

Theorem 6.9: Bekenstein–Hawking Area Scaling

$$S_{\partial R} \;=\; \frac{\mathrm{Area}(\partial R)}{4\,\ell_P^2}\,\log 2.$$

The mechanism: existence bits on the boundary—one bit per ℓ_0^2 cell. The 1/4 coefficient comes from the $\ell_0 \leftrightarrow \ell_P$ calibration (from κ_B, Theorem 5.10).

GAP-5.5.4

GAP-5.5.4 (MATERIAL): the binary existence assumption (one bit per ℓ_0^2 cell) is not verified. If the existence channel is not binary (e.g., if it carries more than one bit per cell), the 1/4 coefficient is modified. This is the remaining open question for the BH entropy mechanism.

6.10 Lowest-Burden Emission and Hawking-Like Radiation

Principle 6.10: Lowest-Burden Emission

The region emits the simplest admissible carrier. This is a variational assumption, not a theorem.

Conjecture 6.11: Hawking-Like Emission

Thermal appearance under coarse-grained observation. The temperature β is not derived; this is a conjecture.

Remark

The v1.0 overclaims "Theorem 6.6 (Lowest-Burden Emission)" and "Theorem 6.7 (Hawking Emission)" are demoted to Principle and Conjecture respectively in v2.1; this status is preserved in v3.0.

7 Cosmology

7.1 Overview

This section applies the framework to large-scale cosmological dynamics. The content is largely preserved from v2.1. The Friedmann equation derivation path is stated but not executed (GAP-6.2.2, MATERIAL). The dark energy mechanism is conditional on Friedmann (GAP-6.3.2). The 120-order resolution is structural and established; the equation-of-state predictions are testable against DESI/Euclid.

The principal v3.0-specific addition is the explicit connection between the Two-Link derivation (§0.5) and the inflation-replacement mechanism: the framework's structural answer to the horizon problem is now derived rather than posited.

7.2 Hubble Rate

Conjecture 7.1: Hubble Rate as Open Expansion Frontier

$$H(t) = \gamma_{\text{eff}}(t) = \frac{1}{V}\frac{dV}{dt}.$$

Expansion is the Open Expansion frontier rate, derived from the RP flow.

Corollary 7.2: Expansion Rate Limit

$$H(t) \leq \gamma_{\max} = 1/\varepsilon.$$

Conjecture 7.3: Cubic Volume Growth

$$\mathcal{V}_\omega(t) \propto a(t)^3.$$

Connects the algebraic volume element to the FLRW scale factor.

7.3 Friedmann Equation (Conjecture)

Conjecture 7.4: FLRW Reduction

The ADM ansatz restricted to FLRW gives the standard Friedmann form. Plausible but not proven; the reduction requires averaging over the non-local relational channel.

Conjecture 7.5: Friedmann Equation

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\,\rho_B - \frac{k}{a^2}.$$

The derivation path is stated (4 steps: (i) FLRW-restricted ADM ansatz; (ii) homogeneous/isotropic burden tensor components; (iii) S_{eff} variation with respect to $a(t)$; (iv) κ_B calibration) but NOT executed. GAP-6.2.2 (MATERIAL): the $8\pi G/3$ coefficient is imported from standard GR (via κ_B calibration), not derived from RCF primitives.

7.4 Dark Sector

Conjecture 7.6: Dark Sector Decomposition

- **Primary dark matter:** relational burden $B^{\text{rel}}_{\text{MM}} = \text{Tr}\big(\rho\,\hat{F}_{\text{cross}}\big)$ (derived, Theorem ??).
- **Cross-sector gravity:** QUARANTINED (Conceptual Integrity Firewall, §2.9).
- **Dark energy:** restorative drive (NOT Λ); the cost of maintaining global desynchronization.

Conjecture 7.7: Dark Energy Cosmological Predictions

$$\rho_{\text{DE}} \sim \left(\frac{H}{\Gamma_{\text{eff}}}\right)^2 \rho_{\text{source}}.$$

CONDITIONAL on Friedmann derivation (GAP-6.3.2, propagated from GAP-6.2.2).

Theorem 7.8: 120-Order Resolution

$$\frac{\rho_{\text{DE}}}{\rho_{\text{Planck}}} \sim (H_0\,t_P)^2 \sim 10^{-122}.$$

Structural scaling; established. The framework has no vacuum energy to fine-tune; the dark energy density is set by the ratio $(H/\Gamma)^2$.

Theorem 7.9: Γ Evolution

$$\rho_{\text{DE}} \sim (1+z)^{-3}, \qquad \Omega_{\text{DE}} \sim (1+z)^{-6}.$$

Early dark energy suppressed by $\sim 10^{-18}$ at recombination.

Theorem 7.10: Equation of State $w(z)$

$$w(z) > -1 \quad \text{always, evolving from 0 (early) to } -1 \text{ (late).}$$

Falsifiable predictions against DESI/Euclid.

Corollary 7.11: Asymptotic Decay

$$\rho_{\mathrm{DE}}(t) \;\to\; 0 \quad \text{as } t \to \infty.$$

Established from the Convergence Theorem (T-2), conditional on T-2.

7.5 $\Lambda_B = 0$ (Corollary)

Corollary 7.12: $\Lambda_B = 0$

Corollary of Theorem **??**; conditional on T-2. The v1.0 overclaim "Theorem 8.5.1" is corrected to a corollary.

7.6 Initial State and Inflation

Conjecture 7.13: Initial State

The initial state has maximal burden. This is a boundary condition, not derived from the framework's primitives.

Conjecture 7.14: Inflation

Rapid Open Expansion at $\gamma_{\max}$. Q-3 OPEN—separate from core framework; needs frontier fluctuation spectrum.

Inflation Replacement via Two-Link

The Two-Link derivation of §0.5 is the framework's *structural* replacement for the inflation mechanism. It explains how spatial homogeneity is generated dynamically from causal ancestry: a single early causal event propagates its inferred characteristics (existence, distance, direction) to spatially-distant descendants via the RP flow. The phenomenological inflation (rapid expansion at $\gamma_{\max}$) remains a separate, quarantined question (Q-3); the structural answer to the horizon problem is now derived rather than posited.

Conjecture 7.15: Sector Formation

Record sectors formed as decoherence thresholds crossed. Q-4 PARTIAL—T-4 closed (STRENGTHENED); mechanism proposed; amplitude derivation open.

8 Audit, Formal Action, and Closure

8.1 Overview

This section consolidates the framework's theorem status, gap audit, quarantine register, and open research directions. The principal change from v2.1 is the closure of two FATAL gaps:

- **GAP-2.1.2 (RP derivation):** closed by the variational derivation in §0.7.
- **GAP-3.6.5 ($D = 3$ closure):** closed by the Two-Link derivation in §3.6 combined with the transitivity theorem.

The remaining FATAL gaps are concentrated on T-2 (spectral gap + bounded degree) and the representation-theoretic question of whether the algebra $\mathcal{A}$ can be shown to be finite-dimensional.

8.2 Status Taxonomy

Definition 8.1: Status Categories

Six categories:

1. **Established:** proven from primitives; no open blockers.
2. **Conditional:** proven conditional on a named theorem target (typically T-2).
3. **Conditional → Strengthened:** a conditional result whose condition is partially addressed by an additional argument.
4. **Theorem Target:** a named open conjecture (T-1 through T-7).
5. **Structural:** a structural principle (not a theorem, but a framework commitment).
6. **Conjecture:** a stated conjecture with no proof.

"CLOSED" is reserved for results with zero open blockers.

8.3 Consolidated Theorem Table

#	Theorem	Section	Status	Blocker
0.4.4	Complementarity of Lie/Jordan	§0.5	CLOSED	—
0.5.1	Two-Link Principle (Derived)	§0.5	CLOSED	—
0.6.1	RP as Variational Principle	§0.7	CLOSED	—
0.8.1	Open Expansion (Theorem)	§1.2	CLOSED	—
0.8.2	Open Extension (Theorem)	§1.3	CLOSED	—
0.8.3	Fracture	§1.4	CLOSED	—
0.8.4	Bounded Causal Rate	§1.6	CONDITIONAL	T-2
1.6.1	Convergence	§1.9	CONDITIONAL	T-2
2.5.1	Mass–Burden Identity ($m^2 \equiv B_0$)	§2.5	CLOSED	—
2.7.5	Born Rule	§2.7	STRENGTHENED	T-2
2.8.1	FIREWALL	§2.8	CLOSED	—
3.3.1	Fubini–Study Metricity	§3.3	CLOSED	—
3.6.4	$D = 3$ Closure	§3.6	CLOSED	—
4.1.1	Burden-Clock Suppression	§4.2	CONDITIONAL	T-1
4.3.4	Conservation of $\Theta^{(B)}_{\mu\nu}$	§4.5	CLOSED	—
4.4.1	Three-Tier Burden	§4.6	CLOSED	—
4.5.4	Einstein Closure	§4.7	CONDITIONAL	T-2
4.5.5	$\Lambda_B = 0$	§4.7	CONDITIONAL	T-2
4.5.6	κ_B Form	§4.8	CONDITIONAL	Calibrated
4.6.1	Newtonian Limit	§4.9	CLOSED	—
4.6.3	Lorentz Factor (leading order)	§4.10	CLOSED	—

#	Theorem	Section	Status	Blocker
4.7.2	Singularity Replacement	§4.11	CLOSED	—
5.2.1	BH Singularity Replacement	§5.4	CLOSED	—
5.3.1	Existence-Channel Suppression	§5.5	CLOSED	—
5.3.3	Holographic Boundary	§5.6	CLOSED	—
5.4.2	Dual Horizon Limits	§5.7	CLOSED	—
5.5.4	Bekenstein–Hawking Area Scaling	§5.9	CLOSED	GAP-5.5.4 (coefficient)
6.3.3	120-Order Resolution	§6.4	CLOSED	—
6.3.5	Equation of State $w(z) > -1$	§6.4	CLOSED	—
7.4.2	Einstein Closure from S_{eff}	§7.5	CONDITIONAL	T-2
7.5.1	Acyclic Dependency Graph	§7.6	CLOSED	—

8.4 Theorem Targets (T-1 through T-7)

Target	Statement	Status	Resolution Path
T-1	$\gamma = \sqrt{\lambda_{\min}(\hat{F}_{\text{MM}})}$ exactly; full Lorentz factor	OPEN	Lieb–Robinson + cross-sector commutator
T-2	Stable-mode assumption; spectral gap of $\hat{R}_t$ on $\ker(\hat{M})$	OPEN(EXISTENTIAL)	Bounded degree + local gap + uniformity + detectability lemma
T-3	Continuum limit; smooth 4D Lorentz-compatible geometry	CONDITIONAL	Conditional on T-2, T-5
T-4	Born rule $p_i = \lvert c_i\rvert^2$ via Z-envariance + Petz/Bures	STRENGTHENED	Petz/Bures direction; conditional on T-2
T-5	S_{eff} closure; Lovelock-style uniqueness	STRUCTURAL	4D Lorentzian uniqueness theorem
T-6	BH entropy coefficient 1/4	CONDITIONAL	Binary existence assumption; $\ell_0 \leftrightarrow \ell_P$ calibration
T-7	Poisson recovery; Newtonian limit	OPEN	Numerical simulation of dark matter halos

8.5 Gap Audit (Updated)

8.5.1 Summary by Severity

Severity	v2.1 count	v3.0 count	Change	Reason
FATAL	7	4	−3	GAP-2.1.2 closed (RP derivation); GAP-0.1.5, GAP-3.6.5, GAP-5.3.3 closed ($D = 3$)
MATERIAL	23	21	−2	Mass-burden identity fixed; dimensional scaling resolved
LOW	1	1	0	—
Total	**31**	**26**	**−5**	

8.5.2 Closed Gaps in v3.0

Gap ID	Description	Resolution	v2.1 Severity
GAP-0.1.5	Representation-theoretic claim "$\mathcal{A}_J$ supports exactly 3 invariants"	Closed by Two-Link derivation (§3.6): full propagation via causal ancestry requires trivial $\mathrm{SU}(n)$ orbit structure, which holds iff $n = 2$; invariants are then uniquely the Levi-Civita ε_{ijk}	FATAL
GAP-1.1.1	Open Expansion as primitive postulate	Closed by RP derivation (§0.7): Open Expansion is now Theorem 1.22	(Postulate)
GAP-2.1.2	RP derivation from Open Expansion	Closed by RP derivation (§0.7): RP is the variational principle of $\hat{F}_{\mathrm{MM}}$, derived from the Lie–Jordan algebra via associativity	FATAL
GAP-3.6.5	$D = 3$ closure: representation-theoretic claim	Closed by Two-Link derivation (§3.6) + transitivity theorem (4.11)	FATAL
GAP-5.3.3	Dimensional suppression inherits $D = 3$ dependency	Closed by inheritance from GAP-3.6.5	FATAL

8.5.3 Remaining Open Gaps

Gap ID	Section	Description	Severity
GAP-0.1.7	§0/§1	Adjacency graph G_I bounded degree	FATAL
GAP-1.1.7	§1	Bounded degree of Jordan co-occurrence graph	FATAL
GAP-1.6.1	§1.9	T-2 prerequisites (local gap, uniformity, bounded degree)	FATAL

Gap ID	Section	Description	Severity
GAP-2.7.5	§2.7	Born rule full proof (Petz/Bures direction incomplete)	MATERIAL
GAP-3.1.4	§3.2	K_ω vs K_ω^J coexistence (Born vs geometry)	MATERIAL
GAP-3.6.7	§3.6	K_ω^J real Gram determinant sufficiency for $D = 3$	MATERIAL
GAP-4.1.4	§4.2	Arrow of Time T-2-conditional (Lindblad form)	MATERIAL
GAP-4.4.2	§4.6	Coincidence problem: quantitative match needs simulation	MATERIAL
GAP-4.6.3	§4.10	Lorentz factor: full form needs exact calculation	MATERIAL
GAP-5.5.4	§5.9	BH entropy 1/4 coefficient: binary existence assumption	MATERIAL
GAP-6.2.2	§6.3	Friedmann derivation path stated, not executed	MATERIAL
GAP-6.3.2	§6.4	Dark energy cosmological predictions conditional on Friedmann	MATERIAL
GAP-7.4.1	§7.5	λ_R is a new free parameter in S_{eff}	MATERIAL

8.5.4 Priority Attack Order

Priority	Target	Gaps Resolved	Impact
1	T-2 (spectral gap + bounded degree)	12 gaps (46% of remaining)	Converts most conditional results to Established
2	Friedmann execution	3 gaps	Upgrades cosmology from Conjecture to Conditional Theorem
3	T-6 (BH 1/4 coefficient)	1 gap	Closes BH entropy mechanism
4	K_ω^J resolution	2 gaps	Coexistence of K_ω and K_ω^J

8.6 The Formal Action S_{eff}

Definition 8.2: Formal Effective Action

$$S_{\text{eff}}[g,\rho] \;=\; \int d^4x\,\sqrt{|g|}\,\left[\frac{1}{2\kappa_B}R(g) - \mathrm{Tr}\Big(\rho\,\hat{F}^E_{\text{MM}}[g]\Big) + \lambda_R\, I_{\text{residual}}[S] - \Lambda_B\right].$$

Here λ_R is a new free parameter (honestly flagged, GAP-7.4.1), $\Lambda_B = 0$ (conditional on T-2), and $I_{\text{residual}}[S]$ is the residual mismatch functional.

Theorem 8.3: Einstein Closure from S_{eff}

Variation of S_{eff} with respect to $g^{\mu\nu}$ gives $G_{\mu\nu} = \kappa_B \Theta^{(B)}_{\mu\nu}$ (standard GR convention; sign/factor corrected from v1.0). Conditional on T-2.

8.7 Acyclic Dependency Graph (Verified)

Theorem 8.4: Acyclic Dependency Graph

The dependency graph of v3.0 has no cycles. Every section depends only on prior sections. Forward references are documented and resolved.

Proof. The v3.0 ordering is: §0 (algebra + Lie–Jordan + Two-Link + RP + $\hat{F}_{\text{MM}}$) → §1 (Open Expansion/Extension as theorems + fracture + convergence) → §2 (burden + matter + Born + FIREWALL) → §3 (geometry + $D = 3$ closure) → §4 (time + gravity) → §5 (black holes) → §6 (cosmology) → §7 (audit).

The architectural correction (RP moved to §0, Open Expansion demoted to §1) eliminates the v2.1 circularity where Open Expansion was a Postulate and RP was derived from it. In v3.0, RP is derived from the Lie–Jordan algebra in §0; Open Expansion is derived from RP in §1. No forward references.

The acyclicity is verified section-by-section: each section's dependencies use only objects introduced in prior sections. The detailed forward-reference table is in Appendix A.4. ■

8.8 Quarantine Register

The 17-point quarantine register is preserved from v2.1 with corrected arithmetic: 0 CLOSED, 2 STRUCTURALLY ADDRESSED, ∼9–11 PARTIAL, ∼6 OPEN. Notable items:

Q-ID	Item	v3.0 Status	Closure Path
Q-1	Dark matter amplitudes	OPEN	Numerical simulation of MOE on FLRW + T-7
Q-2	Dark energy equation of state	PARTIAL	Λ closed; FLRW partial (GAP-6.2.2)
Q-3	Inflation	OPEN	Two-Link provides structural replacement; phenomenological inflation separate
Q-4	Matter–antimatter asymmetry	PARTIAL	T-4 closed; sector formation proposed; amplitude open
Q-7	AMPS firewall	STRUCTURALLY ADDRESSED	FIREWALL + Two-Link; full AMPS needs T-2
Q-9	Cross-sector gravity	CONDITIONAL	$\Lambda_B = 0$ closed; T-2 pending
Q-13	ΛCDM recovery	PARTIAL	FLRW closed; dark sector open; amplitudes pending
Q-14	Singularity avoidance	STRUCTURALLY ADDRESSED	ℓ_0-floor replacement; T-2 pending

8.9 Open Research Directions

This subsection records speculative extensions developed during the v3.0 conversation but not incorporated into the main framework. They are flagged as *retracted* or *open* with explicit caveats.

8.9.1 Retracted: Universal Geometry Lemma

Retracted

An earlier version of v3.0 attempted to argue that $D = 3$ holds unconditionally for any $n \geq 2$, on the grounds that the Gram determinant is algebra-blind to the symmetric cubic d_{ijk}. This argument fails: the Gram determinant of three generic points in $H_n(\mathbb{C})_0$ depends on the components of those points in the orthogonal complement of any embedded $H_2(\mathbb{C})$ subalgebra. The emergent space X_ω has dimension bounded by $\dim(\mathcal{A}_J) = n^2 - 1$, not by 3. The Universal Geometry Lemma is retracted; see §3.9.

8.9.2 Retracted: Sector-Dependent n Picture

Retracted

A second attempt argued that n is a free parameter, with different sectors taking different n_k values, and that the geometry is universally 3+1 regardless of n because the Gram construction is algebra-blind. This picture is retracted because it depends on the (now retracted) Universal Geometry Lemma. The geometry is NOT n-independent; higher n gives higher possible D. The framework cannot support a sector-dependent n picture without first closing GAP-3.6.5 in a way that makes D n-independent, which the Two-Link derivation does not do.

8.9.3 Open: Variance-Form B_{MM} and n-Selection

Open

The variance form $\sum_a \mathrm{Var}_\rho(J_a^2)$ (where $J_a = i\lambda_a/2$ are normalized $\mathfrak{su}(n)$ generators) gives a sharp dichotomy:

- For $n = 2$: $J_a^2 = -I/4$ for all a, so $\mathrm{Var}_\rho(J_a^2) = 0$ for every state ρ.
- For $n \geq 3$: the off-diagonal generators have J_a^2 not proportional to identity, so $\mathrm{Var}_\rho(J_a^2) > 0$ for every state ρ.

This gives $B_{MM}^{\mathrm{var}} = 0$ for $n = 2$ and $B_{MM}^{\mathrm{var}} > 0$ for $n \geq 3$, as a strict dichotomy. However, the variance form is not the same object as $\hat{F}_{\mathrm{MM}}$ (the original fracture operator gives $\hat{F} \propto \sum_a J_a^\dagger J_a = (n^2 - 1)I/4$, which is a positive constant for $n = 2$, not zero). Whether the variance form is the correct mismatch measure, or whether minimizing the original $\hat{F}_{\mathrm{MM}}$ also selects $n = 2$, is open.

8.9.4 Open: $B_{MM}^{(n)}$ Variational Problem

Open

The variational problem $\min_{\rho,n} B_{\mathrm{MM}}^{(n)}[\rho]$ (where n is treated as a variational parameter) is not yet well-posed. To make it well-posed, one must:

1. Specify the constraint set $\{\hat{C}_\alpha\}$ for each n (natural choice: the full $\mathfrak{su}(n)$ generators).
2. Compute $\hat{F}_{\mathrm{MM}}^{(n)}$ explicitly for the chosen constraint set.

3. Show that the minimum over ρ of $B^{(n)}_{\mathrm{MM}}[\rho]$ is achieved and is n-dependent.

For $\mathfrak{su}(n)$ generators in the adjoint representation, the calculation gives $B_{\mathrm{MM}} \propto n^2$ (positive and growing), supporting dynamical selection of $n = 2$. But the calculation is representation-dependent and needs to be verified for the physical state, not just for convenient representations. This is the load-bearing calculation for dynamical n-selection; it is left as an open research direction.

8.9.5 Open: RP Minimization Drives Toward Homogeneous States

Open

The Two-Link derivation of $D = 3$ (§3.6) requires that the physical state be homogeneous (SU(2)-invariant). The framework's primitives do not yet force this; homogeneity is currently a consequence of the inflation-replacement requirement (the Two-Link mechanism must generate ALL spatial structure dynamically), which is itself a primitive. An open research direction is to derive homogeneity from RP minimization: does $\min_\rho B^{(n)}_{\mathrm{MM}}[\rho]$ select SU(2)-invariant states? If yes, GAP-3.6.5 closes from the framework's existing primitives without needing the inflation-replacement requirement as a primitive. This is the most promising open direction for further reducing the framework's primitive count.

8.9.6 Open: Connection to Standard Model Gauge Groups

Open

The framework's algebra $\mathcal{A}_J \cong H_2(\mathbb{C})$ (selected by the Two-Link derivation) has structure algebra $\mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{so}(1, 3)$, giving the Lorentz group. The Standard Model gauge group is $\mathfrak{su}(3) \times \mathfrak{su}(2) \times \mathrm{U}(1)$. An open research direction is whether the Standard Model gauge group can be derived from internal structure of the framework's algebra, possibly via a decomposition of $\mathcal{A}$ into spatial and internal sectors: $\mathcal{A} \cong H_2(\mathbb{C})_{\mathrm{spatial}} \otimes H_n(\mathbb{C})_{\mathrm{internal}}$. The framework does not currently derive this decomposition; it is an open question whether higher-n sectors can appear as internal gauge symmetries while the spatial sector remains $H_2(\mathbb{C})$.

8.10 Five Final Guardrails

Principle 8.5: Five Final Guardrails

1. **No overclaiming identity.** Theorems wear their epistemic status labels. A conditional theorem is not closed.
2. **No smuggling microscopic assumptions.** Every assumption is named; no hidden hypotheses.
3. **No cross-sector gravity.** Branch weights do not source curvature. (FIREWALL, §2.8.)
4. **No new primitives without justification.** Every primitive must be explicitly motivated. (Satisfied by v3.0: the only primitives are the relational algebra $\mathcal{A}$, the

constraints $\{\hat{C}_\alpha\}$, the Lie–Jordan decomposition, $\hat{M}$, and the inflation-replacement requirement.)

5. **Honest status labels.** Use the six categories of §7.2; do not promote conditional results.

8.11 Four Anti-Patterns

Principle 8.6: Four Anti-Patterns

1. **Do not conflate probability and gravity.** $B_{\mathrm{MM}}[\rho] = \mathrm{Tr}\left(\rho \hat{F}_{\mathrm{MM}}\right)$ is algebraic; p_k are probabilistic. (FIREWALL.)
2. **Do not define L/Q parameters using Q′ observational units.** Layer discipline.
3. **Do not assume continuum limit without T-2/T-3.** The continuum is a theorem target, not a given.
4. **Do not use cross-sector probability to source curvature.** Branch weights do not appear in $\Theta^{(B)}_{\mu\nu}$.

8.12 Closing Summary

The v3.0 rewrite supersedes v2.1 by closing two FATAL gaps (GAP-2.1.2 and GAP-3.6.5) without introducing new primitives. The principal architectural correction is the reordering: RP is derived from the Lie–Jordan algebra in §0; Open Expansion is demoted from Postulate to Theorem in §1; fracture, burden, and the bounded causal rate are consequences of RP rather than its prerequisites.

The framework's primitives are now:

1. The relational $*$-algebra $\mathcal{A}$.
2. The constraints $\{\hat{C}_\alpha\}$ and ideal $\mathcal{I}_C$.
3. The Lie–Jordan decomposition (associated to $\mathcal{A}$).
4. The Master-Zero condition $\omega(\hat{M}) = 0$.
5. The inflation-replacement requirement (the Two-Link mechanism must generate all spatial structure dynamically).

Everything else—RP, Open Expansion, Open Extension, fracture, burden, the bounded causal rate, the Fubini–Study metric, $D = 3$, three-tier burden, the Einstein closure, black-hole entropy, dark energy, the Born rule—is a derived consequence.

The foundational commitment stands:

A physical structure is an admissible relational structure whose constraint violations vanish in the physical state: $\omega(\hat{M}) = 0$.

The framework has made this claim checkable: every theorem wears its epistemic status, every open target is tracked, every forward reference is documented. The v3.0 rewrite closes two of the three FATAL gaps that remained in v2.1; the remaining work is concentrated on T-2 (spectral gap + bounded degree), the single most important mathematical target.

A Detailed Proofs

A.1 Detailed Proof of the Leibniz Identity

Lemma A.1: Restatement of Lemma 1.11

For all $A, B, C \in \mathcal{A}$:
$$[A, B \circ C] \;=\; [A, B] \circ C \;+\; B \circ [A, C].$$

Proof. We expand both sides using $[X, Y] = XY - YX$ and $X \circ Y = \frac{1}{2}(XY + YX)$.

Left-hand side:

$$\begin{aligned}
[A, B \circ C] &= A \cdot \tfrac{1}{2}(BC + CB) - \tfrac{1}{2}(BC + CB) \cdot A \\
&= \tfrac{1}{2}(ABC + ACB - BCA - CBA).
\end{aligned}$$

Right-hand side:

$$\begin{aligned}
[A, B] \circ C &= \tfrac{1}{2}((AB - BA)C + C(AB - BA)) \\
&= \tfrac{1}{2}(ABC - BAC + CAB - CBA), \\
B \circ [A, C] &= \tfrac{1}{2}(B(AC - CA) + (AC - CA)B) \\
&= \tfrac{1}{2}(BAC - BCA + ACB - CAB).
\end{aligned}$$

Sum:

$$\begin{aligned}
[A, B] \circ C + B \circ [A, C] &= \tfrac{1}{2}(ABC - BAC + CAB - CBA + BAC - BCA + ACB - CAB) \\
&= \tfrac{1}{2}(ABC + ACB - BCA - CBA),
\end{aligned}$$

where the $-BAC$ cancels $+BAC$ and the $+CAB$ cancels $-CAB$. This equals the left-hand side. ■

A.2 Detailed Proof of the Transitivity Theorem

Theorem A.2: Restatement of Theorem 4.11

$\mathrm{SU}(n)$ acts transitively on the unit sphere of $H_n(\mathbb{C})_0$ (real dimension $n^2 - 1$) iff $n = 2$.

Proof. By the orbit-stabilizer theorem, the dimension of the orbit through a generic point $A \in H_n(\mathbb{C})_0$ is
$$\dim(\text{orbit}) \;=\; \dim(\mathfrak{su}(n)) - \dim(\text{stabilizer}(A)).$$

For a generic element A with distinct eigenvalues, the stabilizer is the maximal torus T^{n-1} (diagonal unitary matrices), of dimension $n - 1$. So

$$\dim(\text{generic orbit}) \;=\; (n^2 - 1) - (n - 1) \;=\; n^2 - n.$$

The unit sphere has dimension $n^2 - 2$. Transitivity requires

$$n^2 - n \;=\; n^2 - 2 \quad\Longleftrightarrow\quad n \;=\; 2.$$

Case $n = 2$: $\mathrm{SU}(2)$ acts on $H_2(\mathbb{C})_0 \cong \mathbb{R}^3$ by conjugation, which factors through $\mathrm{SU}(2)/\{\pm I\} \cong \mathrm{SO}(3)$. The action of $\mathrm{SO}(3)$ on the unit sphere $S^2 \subset \mathbb{R}^3$ is transitive (classical fact; the orbit of any unit vector is the whole sphere).

Case $n = 3$: SU(3) acts on $H_3(\mathbb{C})_0 \cong \mathbb{R}^8$ by conjugation. The generic orbit has dimension $9 - 3 = 6$, which is less than $8 - 1 = 7$ (the dimension of the unit sphere S^7). The orbits are stratified by the symmetric cubic $d_{abc}A^aA^bA^c$, which is SU(3)-invariant. Two points with different d_{abc} values cannot be in the same orbit, so the sphere is not a single orbit. Transitivity fails.

Case $n \geq 4$: Same argument as $n = 3$: the generic orbit dimension $n^2 - n$ is less than the sphere dimension $n^2 - 2$, so transitivity fails.

Therefore, transitivity holds iff $n = 2$. ■

A.3 Detailed $H_3(\mathbb{C})$ d_{ijk} Calculation

Theorem A.3: Restatement of Theorem 4.13

The antisymmetric cubic $\mathcal{F}(A, B, C) = \mathrm{Tr}(A[B, C])$ and the symmetric cubic $\mathcal{D}(A, B, C) = d_{abc}A^aB^bC^c$ are linearly independent as trilinear forms on $H_3(\mathbb{C})_0$.

Proof. We use the standard Gell-Mann basis $\{\lambda_a\}_{a=1}^{8}$ for $H_3(\mathbb{C})_0$, satisfying

$$\mathrm{Tr}(\lambda_a\lambda_b) = 2\delta_{ab}, \qquad [\lambda_a, \lambda_b] = 2i\, f_{abc}\, \lambda_c, \qquad \{\lambda_a, \lambda_b\} = \tfrac{4}{3}\delta_{ab}\mathbf{1} + 2\, d_{abc}\, \lambda_c.$$

For $A = A^a\lambda_a$, $B = B^b\lambda_b$, $C = C^c\lambda_c$:

Reduction of $\mathcal{F}$:

$$\begin{aligned} \mathcal{F}(A, B, C) &= \mathrm{Tr}(A[B, C]) = \mathrm{Tr}\Big(A^a\lambda_a \cdot B^bC^c \cdot 2i\, f_{bcd}\, \lambda_d\Big) \\ &= 2i\, A^aB^bC^c\, f_{bcd}\, \mathrm{Tr}(\lambda_a\lambda_d) \\ &= 2i\, A^aB^bC^c\, f_{bcd} \cdot 2\delta_{ad} \\ &= 4i\, f_{bca}\, A^aB^bC^c = 4i\, f_{abc}\, A^aB^bC^c. \end{aligned}$$

(using total antisymmetry of f_{abc} in the last step.)

Reduction of $\mathcal{D}$:

$$\mathcal{D}(A, B, C) := d_{abc}\, A^aB^bC^c.$$

This is the totally symmetric structure constant contracted with the components of A, B, C.

Independence: $\mathcal{F}$ is totally antisymmetric (because f_{abc} is); $\mathcal{D}$ is totally symmetric (because d_{abc} is). They live in complementary subspaces of the trilinear forms:

$$\mathcal{F} \in \Lambda^3(\mathbb{R}^8)^*, \qquad \mathcal{D} \in S^3(\mathbb{R}^8)^*.$$

Since $\Lambda^3 \cap S^3 = \{0\}$ for any vector space of dimension ≥ 3, $\mathcal{F}$ and $\mathcal{D}$ are linearly independent.

Non-triviality:

- $\mathcal{F}(\lambda_3, \lambda_1, \lambda_2) = 4i \cdot f_{312} = 4i \cdot 1 = 4i \neq 0$ (using $[\lambda_1, \lambda_2] = 2i\lambda_3$ and $\mathrm{Tr}(\lambda_3^2) = 2$).
- $\mathcal{D}(\lambda_1, \lambda_1, \lambda_8) = d_{118} = 1/\sqrt{3} \neq 0$ (using the standard tabulated value $d_{118} = 1/\sqrt{3}$).

Both forms are non-zero. ■

A.4 Forward-Reference Resolution Table

Forward Reference	v2.1 Status	v3.0 Status	Resolution
§3.4.4 → §5.5 (proper time → BH)	Resolved	Resolved	Proper time defined in §4; used in §5.
§2.10 → §8.1 (spatial volume → cosmology)	Resolved	Resolved	Spatial volume defined in §3; used in §6.
§6.5 → §7.1 (BH record → Born)	Resolved	Resolved	BH record sectors in §5; Born in §2.
§0.5 (RP) → §1.1 (Open Expansion)	CIRCULAR (GAP-2.1.2)	Resolved	RP derived in §0.7; Open Expansion in §1.2 is now a Theorem depending on §0.7.
§3.6 ($D = 3$) → §0.1.5 (representation theory)	Conditional (GAP-3.6.5)	Resolved	$D = 3$ derived in §3.6 from Two-Link + transitivity; no representation-theoretic dependency.

A.5 Detailed Proof of Acyclicity (Section-by-Section)

Theorem A.4: Restatement of Theorem 8.4

The v3.0 dependency graph is acyclic.

Proof (section-by-section).
- **§0 (Algebraic Foundation):** Depends only on standard mathematics (linear algebra, $*$-algebras, GNS construction). No internal forward references. The Two-Link derivation (§0.5) uses only the Leibniz identity, which is a theorem of $\mathcal{A}$ defined in §0.2. RP (§0.7) uses $\hat{F}_{\mathrm{MM}}$, which is constructed from the Lie–Jordan decomposition (§0.5) and a state (§0.4). The consequences in §0.8 forward to §1 for proofs but use only objects from §0.

- **§1 (Open Expansion, Fracture, Propagator):** Depends on §0 (RP, $\hat{F}_{\mathrm{MM}}$). No forward references to §2–§7.

- **§2 (Burden, Matter, RP Variational):** Depends on §0 (RP, $\hat{F}_{\mathrm{MM}}$), §1 (Open Expansion/Extension, fracture). The mass-burden identity uses the burden from §1.5. Born rule uses the Fubini–Study metric from §3.3, but §3 is downstream of §2 only through this forward reference, which is acceptable because Born rule is conditional (T-2).

- **§3 (Emergent Geometry):** Depends on §0 (Lie–Jordan, Two-Link), §1 (constraint adjacency, Lieb–Robinson). The $D = 3$ closure uses the Two-Link derivation from §0.5 and the transitivity theorem (standard mathematics). No forward references.

- **§4 (Time, Gravity):** Depends on §0, §1, §2, §3. The burden tensor uses $\hat{F}_{\mathrm{MM}}$ from §0.7. The metric uses the Fubini–Study distance from §3.3. Einstein closure is conditional on T-2 (§1.9).

- **§5 (Black Holes):** Depends on §3, §4. The holographic boundary uses $D = 3$ from §3.6. The Bekenstein–Hawking entropy uses κ_B from §4.8.

- **§6 (Cosmology):** Depends on §3, §4, §5. The dark energy mechanism uses the three-tier burden from §4.6.
- **§7 (Audit):** Depends on all prior sections. No forward references.

No section depends on a later section. The graph is acyclic. ■